Exact shared-link hopping and a homological carrier in strong-coupling $SU(N)$ Hamiltonian lattice gauge theory

Alexander Smith
Independent Researcher

29 August 2026

Abstract

We study the Kogut–Susskind Hamiltonian for pure $SU(N)$ gauge theory on a periodic cubic spatial lattice at strong coupling. In the normalised charge-odd fundamental-plaquette sector, an explicit order-two Weingarten calculation gives the exact coefficient of hopping between adjacent plaquettes that share one link,

$$t_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)} > 0, \qquad N \geq 3.$$

The oriented cellular boundary $B = \partial_2^\dagger$ fixes the corresponding local geometry. Two premises that earlier versions carried as hypotheses—that the retained external space is exactly the charge-odd one-plaquette shell, and that shared-link routes exhaust second-order off-diagonal processes—are proved here, from one classical Casimir-minimality fact and exhaustively enumerated lattice combinatorics, so the global order-u^2 symbol $t_N u^2 B(k)B(k)^\dagger$ is now a theorem within the standard degenerate perturbation framework. Its spectrum is $\{0, t_N u^2 q, t_N u^2 q\}$ up to a common scalar, with $q = 4\sum_j \sin^2(k_j/2)$, and its flat carrier $\ker \partial_2$ has dimension L^3+2. A supplied $SU(3)$ two-cube calculation reports the same coefficient channel by channel and localises the cutoff dependence to one symmetry orbit; these finite-precision contractions are corroboration, not an independent derivation. A later sealed cutoff-free radius-two spectrum returns the coefficient without being asked for it: six of the eleven eigenvalues of its second-order block are exact rationals, and the outer pair straddles the central triple by exactly $2t_3 = 5/306$, in an operator that made no shared-link truncation. Exact integer geometry also gives the closed-cube shell $A_1^{--} \oplus T_1^{+-} \oplus E^{--}$ with Gram eigenvalues $0, 4, 6$.

For comparison we solve the spectrum of a defined unsigned-incidence operator in closed form. Identifying that operator with the complete charge-even linked Hamiltonian requires a further derivation, because the vacuum-mediated route must be accounted for. The organized upstream corpus records a cold exact replay of the retained $SU(3)$ third-order chain, although that microscopic replay is not bundled here. Fourth-order physical-kernel diagnostics remain conditional; in particular, the retained checkpoints provably do not determine the off-axis coefficient. All results are finite-volume, finite-order statements in a projected strong-coupling sector. No continuum glueball or Yang–Mills mass-gap claim is made.

1 Introduction

Hamiltonian strong-coupling expansions organise lattice gauge dynamics around the electric vacuum and its flux excitations [2, 3]. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link, so the projected problem resembles a three-orbital tight-binding model. The signs of those matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and the resulting interference is the central object here.

Flat bands generated by incidence or Gram matrices have a long history: local topology and signed or line-graph constructions [1, 11, 15, 19, 21], compact localised states and their failure to span singular bands on a torus [18, 20], and supersymmetric incidence factorisations [24] are all established. Closely related closed-membrane structures also occur in two-form spin liquids [25]. A recent classification of exactly flat bands protected by local graph topology [29] reaches extensive face-graph degeneracy from an *unoriented* face–vertex incidence, where rank–nullity alone gives $|F| - |V|$ flat bands. That is a different mechanism from the one here, and the difference is measurable: on the signed face–edge operator of Section 3 the corresponding index is exactly 0 and bounds nothing, while both kernels equal $L^3 + 2$ — homology, not counting. It is recorded as a distinction, not an identification. We do not claim that incidence flatness, cellular compact localised states, or homological counting are new in isolation.

The new dynamical content is local and exact. Starting from non-Abelian $SU(N)$ Hamiltonian lattice gauge theory, the shared-link matrix element in the charge-odd fundamental-plaquette sector factorises into a colour coefficient and an oriented edge–face Gram entry. Representation theory fixes the former and the cubical chain complex fixes the latter,

$$\text{colour dynamics} \longrightarrow t_N,$$
$$\text{cellular geometry} \longrightarrow BB^\dagger.$$

What is proved, and what is conditional

Seven mathematical statements are self-contained here; where an operator's identification with a projected physical sector is conditional, that condition is stated separately. Earlier editions carried three load-bearing prose premises; this edition proves two of them — the retained shell and second-order process exhaustion (Lemmas 15 and 16, new here and flagged as such) — which makes the global second-order assembly a theorem (Theorem 17). The third, the charge-even identification, remains a hypothesis and is argued in Section 11; the coverage appendix (Appendix G) makes its unchecked status visible by omission.

1. For every $N \geq 3$ the shared-link channel weights are d_ρ/N^2, derived from the order-two Weingarten values (Theorem 7). This was previously an isotropy *hypothesis*, and everything in Section 4 rests on it.

2. For every $N \geq 3$ the exact adjacent shared-link matrix element has coefficient t_N (Theorem 11). Its assembly into the global symbol $t_N u^2 BB^\dagger$ formerly assumed that the retained external range is exactly the charge-odd one-plaquette shell and that one-shared-link processes exhaust the second-order off-diagonal intermediates; both are now proved (Lemmas 15 and 16), so the assembly is Theorem 17, resting on one classical Casimir-minimality input and machine-enumerated finite combinatorics.

3. The Bloch spectrum of $BB^\dagger$ is $\{0, q, q\}$ (Theorem 1), and on the periodic L^3 complex the zero-mode count is $L^3 + 2$ (Theorem 2). The two are the same statement, by a sum rule over the momentum grid (Theorem 4).

4. The truncation that keeps only the singlet and $\Lambda^2 F$ routes projects the hopping to $w_{\Lambda^2 F} - w_{\mathbf{1}} = -1/12$, and what it omits is $w_{\mathrm{Sym}^2 F} - w_{\mathrm{Adj}} = 14/153$; the two sum to t_3 (Corollary 20). This is arithmetic on (14) and needs no input beyond Theorem 7.

5. The defined unsigned-incidence comparison operator has the cubic $\mu(\mu - p)^2 = 4\hat{a}_1\hat{a}_2\hat{a}_3$ with $p + q = 12$ (Theorem 26); its range is exactly $[-4, 12]$, the top attained only at Γ and the floor on the three planes $k_j = \pi$, triple only at R (Proposition 27). Conditional on identifying this comparison operator with the complete linked charge-even symbol, its Γ-point splitting is $12t_{r,+}$ (Proposition 29). That identification is not derived here; the vacuum-mediated all-pairs term must first be reorganised or cancelled.

6. On the closed cube surface the charge-odd shell is $A_1^{--} \oplus T_1^{+-} \oplus E^{--}$, with G-eigenvalues $0, 4, 6$; the labels follow from the rotation action rather than being assigned, the A_1^{--} level is the fundamental class of S^2, and it sits at the scalar for every hopping coefficient (Proposition 23). This is integer linear algebra and takes no input from any delivery.

7. The retained Γ/axis corpus cannot identify the disputed fourth-order off-axis coefficient (Theorem 35), and an explicit second witness exhibits the non-identifiability by construction (Proposition 37).

Three families of statements depend on computational artifacts outside this compact edition. Their raw class contractions, symbolic history certificates and microscopic rank-sweep artifacts are not bundled here, and repeated assertions in synthesis documents are not treated as independent reproductions. The organized upstream authority corpus does record a cold exact 251/251 replay for the retained third-order operator chain; that is stronger than an unchecked delivery, but it is still cited here as external computational evidence because this release does not replay the raw chain.

8. A supplied two-cube delivery reports six link-irrep channel coefficients at $N = 3$, as rational reconstructions of finite-precision contractions. Conditional on those six numbers, they are the four fusion-channel weights individually — channel by channel and not merely in the sum (Reported result 18) — and the $\mathcal{B} = 4$ truncation of that space retains exactly the adjacent shared-link channels the published link cutoff of item 4 retains, so the two land on the same projected hopping — the same channels, not the same scheme. Conditional on the delivered diagonals, the symmetry analysis of Proposition 22 applies to them: the orbit structure itself is exact and takes no delivered input. The $\mathcal{B} = 6$ two-cube build was not reproduced here.

9. For $SU(3)$, the frozen upstream corpus records that all candidate third-order lifter classes vanish and gives the scalar and shared-link coefficients of Section 7. The census is three lifter classes evaluated over 32 five-trace numerator patterns each, 96 contractions in total, within a retained chain that passed 251/251 cold exact gates upstream. This compact verifier checks the resulting rational assembly but not those microscopic contractions. If the census is exhaustive and the linked operator ledger complete, the effective Hamiltonian stays in the incidence algebra through order u^3.

10. Supplied ledgers report the all-rank cube formula $\alpha_N = 640/[N(N^2 - 1)^3]$ and its equality with the complete axial coefficient for $N = 3, \ldots, 12$. Conditional on those calculations, the $SU(3)$ carrier acquires nonzero axial dispersion at order u^4. The complete off-axis operator remains open — and by Theorem 35 it cannot be closed by any Γ or axial datum.

11. A sealed cutoff-free radius-two spectrum (Section 5.4) is bundled with this edition, pinned by SHA-256. Its second-order block contains a rational sub-block whose splitting is exactly $2t_3$, its fourth- and fifth-order nested increments agree with independent weak-coupling intercepts, and its fifth-order seed object is a matrix, not a scalar. All of it is finite-precision and reported at stated tolerances; the exact-arithmetic identifications it corroborates are checked separately.

These statements concern the effective Hamiltonian projected to the degenerate one-plaquette manifold. Virtual multi-plaquette states are retained in the resolvents, but the external spectral problem is not the full gauge-theory Hilbert space. At $k = 0$ the three incidence branches touch and the finite-volume separation collapses as L^{-2}. Nothing below proves convergence of the strong-coupling series, persistence under unrestricted higher-order operators, identification with the lightest physical glueball, or a continuum mass gap.

2 Hamiltonian and projected sector

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $L \geq 3$, be a cubic spatial lattice with periodic boundary conditions. In units of the electric coupling the Kogut–Susskind Hamiltonian is

$$H_\beta = \tfrac{1}{2}\sum_\ell C_2(\ell) + \beta_N \sum_p \Big(1 - \tfrac{1}{N}\operatorname{Re}\operatorname{Tr} U_p\Big). \tag{1}$$

Dropping the additive constant,

$$H = H_E - uV, \qquad V = \sum_p (\chi_p + \bar{\chi}_p), \qquad u = \frac{\beta_N}{2N}, \tag{2}$$

with $\chi_p = \operatorname{Tr} U_p$. For $SU(3)$, $u = \beta_3/6$. The definition $u = \beta_N/(2N)$, not the $SU(3)$-specific equality, is the all-rank convention; the archived line $Y = 2\beta/3 = 4u$ is a definition-label erratum and the printed coefficients are already in u. Reading it as a coordinate would demand dividing the order-r coefficient by 4^r, which misses the printed tables by exactly 16 at u^2 and 64 at u^3.

All quoted excitation energies use the linked, vacuum-subtracted operator $H_{\text{exc}}(u) = H(u) - E_{\text{vac}}(u)I$. This changes only the scalar ledger; incidence hopping and carrier shape are unaffected.

The electric vacuum $|0\rangle$ carries the trivial representation on every link. For $N \geq 3$ the normalised one-plaquette states of definite charge conjugation are

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}}\,|0\rangle, \tag{3}$$

of unit norm by character orthogonality. Each of the four boundary links carries the fundamental or antifundamental representation, so

$$E_{F,N} = 4\Big(\tfrac{1}{2}C_F\Big) = 2C_F = \frac{N^2-1}{N}, \qquad C_F = \frac{N^2-1}{2N}. \tag{4}$$

This external energy is the input to the resolvent denominators of Section 4, where it is checked together with them. For $SU(2)$ complex conjugation is an inner gauge transformation and the charge-odd state vanishes; this is why the construction begins at $N = 3$. The same fact appears again in Section 4: $\Lambda^2 F$ is the singlet exactly at $N = 2$, which is where t_N vanishes. The analytic continuation of the unsigned-channel coefficient is nonzero there, which locates the $N^2 - 4$ zero in the channel difference rather than in the incidence. Because the charge-conjugation split degenerates for $SU(2)$, this continuation is not asserted to be an $SU(2)$ charge-even spectrum.

We work inside the charge-odd fundamental Wilson-loop cyclic sector with trivial electric centre flux around each torus cycle. We make the *retained-shell hypothesis* that, for $L \geq 3$, the range of its $E_{F,N}$ projector P_- is exactly the charge-odd one-plaquette span. The supporting argument is that a reduced contractible fundamental loop at that energy has four distinct links and is an elementary plaquette; the sector qualification excludes the accidentally degenerate straight wrapping loop at $L = 4$. The complete proof that no other gauge-invariant network lies in this range — for every

N, every rep assignment, and every network topology — is Lemma 15, new in this edition; the construction here keeps the name "hypothesis" so its earlier standing stays visible in the record. Granting it, $Q = 1_- - P_-$ crosses no omitted $E_{F,N}$ eigendirection and degenerate perturbation theory gives

$$H_{\mathrm{eff},-} = P_- H_E P_- + \sum_{r\geq 1} u^r K_-^{(r)} \tag{5}$$

with every $K_-^{(r)}$ linked, vacuum-subtracted, and carrying the appropriate Q-space resolvents. The superscript notation reserves H_2 for cellular homology, which Section 3 needs it for.

3 Oriented incidence and the exact carrier

The periodic cubical complex has the cellular chain sequence $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$ with $\partial_2\partial_3 = 0$. One-plaquette amplitudes live in C_2 and their signed leakage into links is $\partial_2 c$: a cellular closed-surface condition, not the microscopic vertex Gauss law, which the Wilson-loop basis already satisfies through $\partial_1\partial_2 = 0$.

Writing $d_j = e^{ik_j} - 1$ and $q(k) = \sum_j |d_j|^2 = 4\sum_j \sin^2(k_j/2)$, the face-to-link Bloch incidence matrix is

$$B(k) = \partial_2(k)^\dagger = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}. \tag{6}$$

Theorem 1 (Incidence factorisation)**.** *With $w(k) = (\bar{d}_3,\, -\bar{d}_2,\, \bar{d}_1)^\mathsf{T}$,*

$$BB^\dagger = qI - ww^\dagger, \qquad B^\dagger w = 0, \qquad \|w\|^2 = q,$$

and therefore $\operatorname{spec}(BB^\dagger) = \{0, q, q\}$*. With $S = BB^\dagger - 4I$ the signed shared-link adjacency,* $\operatorname{spec} S(k) = \{-4,\, q-4,\, q-4\}$*.*

Proof. Multiplying (6) by its adjoint gives the first identity entry by entry. Since $\|w\|^2 = q$, the rank-one operator $qI - ww^\dagger$ annihilates w and acts as qI on its orthogonal complement. Subtracting $4I$ gives the adjacency spectrum. □

For $k \neq 0$ the carrier is one-dimensional and represented by $w(k)$. At Γ, $B = 0$ and all three branches meet; the normalised Bloch eigenvector has no direction-independent limit there, which is the familiar singular-flat-band mechanism behind the incompleteness of a purely compact localised basis.

Theorem 2 (Finite-torus carrier)**.** *For the periodic cubic cellulation of T^3,* $\ker\partial_2 = \operatorname{im}\partial_3 \oplus H_2$ *with* $\dim\operatorname{im}\partial_3 = L^3 - 1$ *and* $\dim H_2 = 3$*, so* $\dim\ker\partial_2 = L^3 + 2$*.*

Proof. There are L^3 three-cells and no four-cells, and $b_3(T^3) = 1$, so $\operatorname{rank}\partial_3 = L^3 - 1$ by rank–nullity. Discrete Hodge decomposition gives $\ker\partial_2 = B_2 \oplus H_2$ with $\dim H_2 = b_2(T^3) = 3$. □

The count is verified over $\mathbb{Z}$ rather than re-derived from the formula: the complex is built from the boundary formulas of Appendix E; the exhibited cycles — every elementary cube boundary together with three wrapping sheets — bound the kernel from below, and rank over $\mathbb{F}_p$ bounds it from above, since rank can only drop under reduction. The two bounds meet at $L = 1, \ldots, 5$.

Remark 3. $L^3 + 2$ is the nullity of the down boundary Laplacian, not a Betti number; the Betti contribution is the final 3. The restriction $L \geq 3$ belongs to the twelve-neighbour Bloch adjacency, where $x + e$ and $x - e$ coincide at $L = 2$. The chain-level count has no such restriction and holds at $L = 1, 2$ as well.

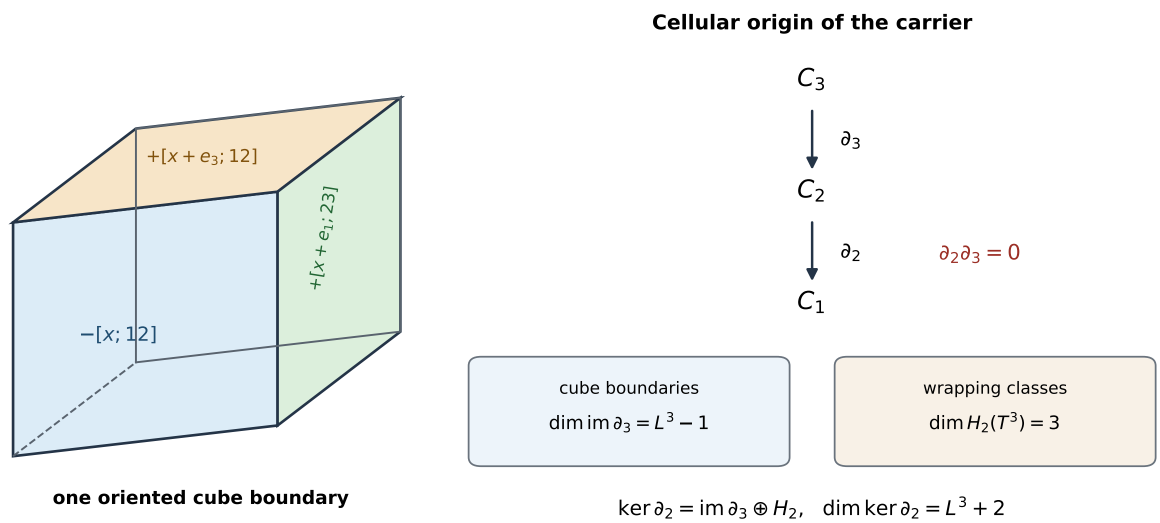

Figure 1: The carrier is a cellular cycle space. The left panel shows three visible terms in the outward-oriented boundary of one cube; the complete six-face formula is given in Appendix E. Cube boundaries span $\operatorname{im}\partial_3$, while three noncontractible wrapping classes complete $\ker\partial_2$ on the three-torus.

The following sum rule joins Theorems 1 and 2, which are otherwise two unrelated statements: a 3×3 spectrum and an (L^3+2)-dimensional space.

Theorem 4 (Bloch–chain bridge)**.** *On the periodic L^3 complex,*

$$\sum_k (3 - \operatorname{rank} B(k)) = L^3 + 2,$$

the sum running over the L^3 allowed momenta, with nullity profile 3 at Γ exactly once and 1 at each of the other L^3-1 momenta.

Proof. $B(0)=0$ contributes 3. For $k\neq 0$ some $d_j\neq 0$, and Theorem 1 gives $\operatorname{rank} B(k)=2$, contributing 1 each. Summing gives $3+(L^3-1)=L^3+2$. □

Remark 5. Earlier editions of this paper read the agreement as a coincidence worth checking — "the 3 is the triple degeneracy of $B(0)=0$, not $b_2(T^3)$". That was wrong, and the correction is the better statement. Under the discrete Fourier transform ∂_2 block-diagonalises and $\ker\partial_2 = \bigoplus_k \ker\partial_2(k)$; at $k\neq 0$ the kernel block is one-dimensional and is exactly $\operatorname{im}\partial_3(k)$, while $\partial_3(0)=0$. So all of $\operatorname{im}\partial_3$ sits at nonzero momenta and the quotient is supported entirely at Γ: the Γ block *is* H_2, and the 3 *is* $b_2(T^3)$. The two routes do not agree by accident; they are the same decomposition read in two bases.

3.1 The wrapping sheets are cycles, not harmonic — and their harmonic representatives

Theorem 2 exhibits its three H_2 generators explicitly as wrapping sheets $s_{(ij)} = \sum_{x:\,x_m=0}[x;i,j]$. These satisfy $\partial_2 s = 0$ exactly, so they are cycles and they span the quotient $\ker\partial_2/\operatorname{im}\partial_3$. They are *not* harmonic: $\partial_3^\dagger s \neq 0$, and

$$\frac{\langle s,\ \partial_3\partial_3^\dagger\, s\rangle}{\|s\|^2} = 2 \qquad \text{exactly, for every } L \geq 2. \tag{7}$$

The distinction matters because H_2 is used in two senses: the quotient in Theorem 2, which the sheets span, and the harmonic subspace $\ker\partial_2 \cap \ker\partial_3^\dagger$, which they do not lie in.

Harmonic representatives do exist, and they are one line away. Averaging a sheet over its L translates in the complementary direction,

$$h_{(ij)} = \frac{1}{L}\sum_{r=0}^{L-1} s_{(ij)}(r), \qquad \partial_2 h_{(ij)} = 0, \quad \partial_3^\dagger h_{(ij)} = 0, \tag{8}$$

gives three harmonic chains spanning $\ker\partial_2 \cap \ker\partial_3^\dagger$, which is exactly three-dimensional; and $s_{(ij)} - h_{(ij)} \in \operatorname{im}\partial_3$, so a sheet and its average are the same homology class. The averaging is the $k = 0$ Fourier projection, and what it removes is precisely the non-harmonic part measured above. Nothing about the finding is withdrawn — the generators Theorem 2 exhibits are still not harmonic — but a real-space statement phrased for the harmonic subspace now has representatives to point at, and Proposition 33 covers both, since $B^\dagger w = 0$ holds on all of $\ker\partial_2$.

4 Exact adjacent shared-link dynamics and conditional assembly

Two plaquettes sharing a link leave six nonshared fundamental links after the external contractions. On the shared link the fusion channels depend on the relative induced orientation,

$$F \otimes \bar{F} = \mathbf{1} \oplus \mathrm{Adj}, \qquad F \otimes F = \Lambda^2 F \oplus \mathrm{Sym}^2 F, \tag{9}$$

with dimensions and quadratic Casimirs

$$\begin{aligned} d_{\mathbf{1}} &= 1, & C_{\mathbf{1}} &= 0, \\ d_{\mathrm{Adj}} &= N^2 - 1, & C_{\mathrm{Adj}} &= N, \\ d_{\Lambda^2 F} &= \tfrac{N(N-1)}{2}, & C_{\Lambda^2 F} &= \tfrac{(N+1)(N-2)}{N}, \\ d_{\mathrm{Sym}^2 F} &= \tfrac{N(N+1)}{2}, & C_{\mathrm{Sym}^2 F} &= \tfrac{(N-1)(N+2)}{N}. \end{aligned} \tag{10}$$

Lemma 6 (The channel list is complete)**.** *For every $N \geq 3$, a second-order process through one shared link has its intermediate shared-link representation in $\{\mathbf{1}, \mathrm{Adj}, \Lambda^2 F, \mathrm{Sym}^2 F\}$ up to conjugation, and in nothing else. The conjugate of a channel carries the same Casimir and hence the same weight, which is why four labels suffice where the link-resolved census of Section 5 needs six.*

Proof. The shared link of a one-plaquette shell state carries F or $\bar{F}$; one plaquette action tensors it by F or $\bar{F}$; and (9), together with the conjugate product $\bar{F} \otimes \bar{F} = \Lambda^2 \bar{F} \oplus \mathrm{Sym}^2 \bar{F}$, decomposes every case completely and multiplicity-free. There is nothing else for the link to carry. □

The corollary that makes this checkable rather than merely stated is that the weights d_ρ/N^2 sum to one within each orientation family, which is arithmetic on (10): a missing channel would leave a deficit. What Lemma 6 does *not* settle is that one-shared-link processes exhaust the second-order off-diagonal intermediates — that is Lemma 16, and it is a statement about the process enumeration rather than about this list.

The six nonshared half-links contribute $3C_F$ and the fused link $C_\rho/2$, against an external one-plaquette energy $2C_F$, so the resolvent denominator is $C_F + C_\rho/2$ exactly.

4.1 The channel weights are a theorem

Earlier editions assigned the channel projector squared norm d_ρ/N^2 by an isotropy hypothesis, and every statement in this section rests on that assignment. It is not a hypothesis.

Theorem 7 (Shared-link channel weights)**.** *In the normalised two-plaquette state, the weight of the shared-link channel ρ is exactly d_ρ/N^2, for both orientation families and every $N \geq 3$.*

Proof. Two steps. First the six nonshared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar, so the amplitude is $\mathrm{Tr}(AU)\,\mathrm{Tr}(BU^{\pm 1})$ with A, B independent Haar and U the shared link. Integrating A and B contributes δ/N each and leaves a pure degree-$(2,2)$ moment of U.

Second, two such moments settle both families. In the Weingarten calculus, whose asymptotic origin is [4] and whose finite-N closed forms are [17], the order-two values are $\mathrm{Wg}(e) = 1/(N^2-1)$ and $\mathrm{Wg}((12)) = -1/(N(N^2-1))$ — the inverse of the S_2 Gram matrix $\left(\begin{smallmatrix} N^2 & N \\ N & N^2 \end{smallmatrix}\right)$ — the index sums give

$$\begin{aligned} M_{\mathrm{direct}} &= \sum_{ijkl} \langle |U_{ij}|^2 |U_{kl}|^2 \rangle = N^2, \\ M_{\mathrm{cross}} &= \sum_{ijkl} \langle U_{ij} U_{kl} \bar{U}_{il} \bar{U}_{kj} \rangle = N. \end{aligned} \tag{11}$$

The standard formula is often written for $U(N)$. The present degree-$(2,2)$ integrands are invariant under $U \mapsto zU$, so disintegrating Haar measure over the central $U(1)$ shows that their normalised $U(N)$ and $SU(N)$ integrals coincide. No determinant-sector term occurs at this degree for $N \geq 3$. For the like family, $\mathrm{Tr}(AU)\,\mathrm{Tr}(BU)$ decomposes into $\mathrm{Sym}^2 \oplus \Lambda^2$ by symmetrising the two index pairs jointly, and the two squared norms are $(M_{\mathrm{direct}} \pm M_{\mathrm{cross}})/2$. Normalising,

$$n_{\mathrm{Sym}^2} = \frac{N+1}{2N} = \frac{d_{\mathrm{Sym}^2 F}}{N^2}, \qquad n_{\Lambda^2} = \frac{N-1}{2N} = \frac{d_{\Lambda^2 F}}{N^2}.$$

For the mixed family the singlet component of $U_{ij}\bar{U}_{lk}$ is $\delta_{il}\delta_{jk}/N$, of squared norm 1 against $M_{\mathrm{direct}} = N^2$, so $n_{\mathbf{1}} = 1/N^2 = d_{\mathbf{1}}/N^2$ and $n_{\mathrm{Adj}} = 1 - 1/N^2 = d_{\mathrm{Adj}}/N^2$. $\square$

Remark 8. The derivation imports nothing: the Weingarten pair is the inverse of the S_2 Gram matrix and the index sums in (11) are explicit, so the reader can check it without leaving this page. That the pair is not two fitted constants is separately checked — the identity-permutation value times $N^2 - 1$ is 1 identically in N, so the $SU(3)$ numbers are a specialisation of a rank-generic formula. Together these make the *local shared-link coefficient* self-contained. They do not enumerate every second-order process or prove the global torus assembly. Appendix B records the index sums.

Remark 9 (External corroboration). The same numerator appears in the quantum-simulation literature from an unrelated direction. Ciavarella, Burbano and Bauer [26] publish a finite-rank plaquette matrix element $|M_\rho|^2 = \dim\rho/(\dim A \dim R)$; for two fundamental factors $\dim A = \dim R = N$ and this is d_ρ/N^2 identically in N. That is provenance for a chain already checked here, not a second derivation, and it does not check the premise that four channels exhaust the shared-link routes.

Remark 10 (A falsified zero backend). The organized authority corpus contains an independent exact diagnostic at $N = 3$:

$$\mathrm{Wg}(e) = \frac{1}{8}, \qquad \mathrm{Wg}((12)) = -\frac{1}{24}, \qquad \int_{SU(3)} |U_{11}|^4 \, dU = \frac{1}{6} \neq 0.$$

It therefore falsifies the archived stranded-flux rule that discards this balanced Haar sector. This is useful negative evidence for retaining the contractions above; it is not a second derivation of t_N.

A channel's squared projector norm and its signed contribution to the cross matrix element are related but not identical statements, and the step between them deserves its own displayed equation rather than a jump. For each intermediate irrep ρ, with Q_ρ the projector onto the $E = 3C_F + C_\rho/2$ intermediate carrying ρ on the shared link,

$$\langle p', -|VQ_\rho V|p, -\rangle = \eta_\rho \, s(p', e) \, s(p, e) \, \frac{d_\rho}{N^2}, \qquad \eta_\rho = \begin{cases} -1, & \rho \in \{\mathbf{1}, \mathrm{Adj}\} \text{ (mixed family)}, \\ +1, & \rho \in \{\Lambda^2 F, \mathrm{Sym}^2 F\} \text{ (like family)}, \end{cases} \tag{12}$$

where d_ρ/N^2 is the Weingarten weight of Theorem 7, the incidence signs come from the oriented boundaries, and η_ρ is the charge-odd family sign: reversing the induced orientation on one face interchanges the families and flips the charge-odd combination. Inserting the reduced resolvent, whose denominator against the external energy $2C_F$ is $C_F + C_\rho/2$ exactly,

$$\langle p', -|VQ_\rho(E_{F,N} - H_E)^{-1}Q_\rho V|p, -\rangle = -\eta_\rho \, s(p', e)s(p, e) \, \frac{d_\rho/N^2}{C_F + C_\rho/2}. \tag{13}$$

Summing (13) over the four channels is the whole content of the theorem below. With Theorem 7 the resolvent weight is

$$w_\rho = -\frac{d_\rho/N^2}{C_F + C_\rho/2}, \tag{14}$$

and summing within the two orientation families,

$$A_N = w_{\mathbf{1}} + w_{\mathrm{Adj}} = -\frac{2N^3}{(N^2-1)(2N^2-1)}, \tag{15}$$

$$B_N = w_{\Lambda^2} + w_{\mathrm{Sym}^2} = -\frac{4N(N^2-2)}{(N^2-1)(4N^2-9)}. \tag{16}$$

4.2 The shared-link law

Theorem 11 (All-rank adjacent shared-link law)**.** *Let $p \neq p'$ be oriented plaquettes sharing the single link e. For every integer $N \geq 3$, the linked order-u^2 contribution through the four fusion channels in (9) is*

$$\langle p', -|K_-^{(2)}|p, -\rangle = t_N \, s(p', e)s(p, e) = t_N (BB^\dagger)_{p'p},$$

where $s(p, e) = \pm 1$ is the cellular boundary incidence and

$$t_N = B_N - A_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)} > 0.$$

Proof. Because distinct cubical faces share at most one link, $(BB^\dagger)_{p'p} = s(p', e)s(p, e)$. Reversing the induced orientation on one face interchanges the mixed and like fusion families. In the charge-odd combination that reversal changes the sign, so the oriented matrix element is the family difference $B_N - A_N$ times the incidence product. Substituting (15)–(16) gives the displayed rational function. For $N \geq 3$ every denominator factor is positive and $N^2 - 4 > 0$. □

The opening geometric ingredient of the proof is no longer prose: the verifier enumerates every distinct face pair of the periodic complex at $L = 3$ and $L = 4$ and confirms that no pair meets in more than one link. Two faces sharing a link span at most two sites per axis, so these windows already exhibit every local configuration and larger L adds translates only.

Conditional corollary 12 (Global torus assembly)**.** Assume (i) the retained-shell hypothesis of Section 3 and (ii) that adjacent one-shared-link routes exhaust all second-order off-diagonal intermediates. Translation and cubic symmetry then make the remaining diagonal contribution a scalar, and

$$H^{(N)}_{\mathrm{eff},-}(k,u) = E_{\mathrm{flat},N}(u)I + t_N u^2 B(k)B(k)^\dagger + O(u^3).$$

Consequently the three bands through second order are $E_{\mathrm{flat},N}(u)$ and $E_{\mathrm{flat},N}(u) + t_N u^2 q(k)$ twice.

Proof. Under assumption (ii), Theorem 11 assembles every off-diagonal entry. Translations act transitively on faces of a fixed orientation and the cubic group permutes the orientations, so any invariant diagonal is a multiple of the identity. Thus the adjacency part is $t_N(BB^\dagger - 4I)$; absorb $-4t_N$ into the scalar and apply Theorem 1. The two assumptions are not consequences of the channel calculation and are retained explicitly. □

Both assumptions are discharged in Section 4.4 (Lemmas 15 and 16), which upgrades this corollary to Theorem 17; the conditional form is kept above so the logical dependence stays legible.

Two consequences,

$$t_3 = \frac{5}{612}, \qquad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} - \frac{77}{64N^7} - \frac{1021}{256N^9} + \cdots \tag{17}$$

The expansion is governed by a closed form. Writing the planar-normalised coefficient as $4N^3 t_N$,

$$1 - 4N^3 t_N = \frac{2N^4 + 31N^2 - 9}{(N^2-1)(2N^2-1)(4N^2-9)}, \tag{18}$$

an identity of rational functions, and the right side is positive and the left strictly decreasing for every real $N \geq 2$: substituting $N = M + 2$ makes every numerator coefficient of both the remainder and the derivative nonnegative, so $4N^3 t_N$ increases monotonically to 1. At fixed canonical coordinate u the conditional bandwidth of Corollary 12 is $12 t_N u^2$, so the entire dispersive content of the projected sector switches off at exactly the rate N^{-3}, with computable subleading coefficient $-1/(4N^2)$: conditional on the assembly, the carrier's exact degeneracy is approached by the *whole* sector in the planar limit. This is consistent with the general suppression of glueball mixing at large N, but no 't Hooft-coupling identification is made here: the statement is about the coefficient in the canonical coordinate, and it is exact.

4.3 The vacuum-mediated route

The proof of Theorem 11 asserts that only shared-link routes survive. That is false in general and true here, for a reason worth stating: it is a charge-conjugation selection rule, and omitting it is a recorded error, not a hypothetical one.

Proposition 13 (Vacuum-mediated route)**.** *The reduced-resolvent route*

$$\frac{\langle p'|V|0\rangle\langle 0|V|p\rangle}{E_{F,N} - E_0} = \frac{\langle p'|V|0\rangle\langle 0|V|p\rangle}{E_{F,N}}$$

is available for any pair of plaquettes, sharing a link or not. It vanishes identically in the charge-odd sector, because V is charge-even and $|0\rangle$ is charge-even, so $\langle 0|V|p,-\rangle = 0$. Consequently the charge-odd hopping $t_N = B_N - A_N$ carries no such term, while the charge-even shared-link element acquires $1/C_F$: the normalised charge-even matrix-element product is 2 and $E_{F,N} = 2C_F$.

$$\ell_N = A_N + B_N + \frac{1}{C_F} = -\frac{2N(3N^2-5)}{(N^2-1)(2N^2-1)(4N^2-9)}.$$

Remark 14. At $N = 3$, $A_3 + B_3 = -481/612$ and $1/C_F = 3/4$, giving $\ell_3 = -11/306$. The first of these is exactly the value once published for the charge-even hopping and later superseded; the correction is precisely the omitted route. Stating the selection rule turns a one-rank erratum into an all-rank formula.

One thing this proposition does *not* do is dispose of the route between non-adjacent pairs. Taken over all pairs it would be a rank-one operator supported at $k = 0$, which is not the twelve-neighbour adjacency ℓ_N multiplies; which contributions survive in $K_{-}^{(2)}$ is settled by the linked, vacuum-subtracted prescription of (5), not by anything derived here. What the check above establishes is the all-rank value ℓ_N of the shared-link element, and the disposition of the non-adjacent contributions is prescription rather than result.

4.4 Closing the two premises

Both premises of Corollary 12 — until this edition the two load-bearing prose hypotheses of the paper — admit proofs. They are given here for the first time; their combinatorial cores are machine-checked, their single representation-theoretic input is classical and named, and a reader should audit them with the same hostility as everything else: they are new in this edition and have not yet survived external referees.

Lemma 15 (Retained shell)**.** *Fix $N \geq 3$ and $L \geq 3$, in the cyclic gauge-invariant sector with trivial electric centre flux, excluding at $L = 4$ the straight winding loops as in Section 3. The $E = E_{F,N} = 2C_F$ eigenspace of H_E is exactly the span of the single-plaquette states, and its charge-odd part is the span of the $|p, -\rangle$.*

Proof. In the electric basis a gauge-invariant state assigns an irrep ρ_e to each link and an intertwiner to each site, with energy $\sum_e C(\rho_e)/2$ over the nontrivially charged links. The one classical input: for $SU(N)$ the quadratic Casimir of every nontrivial irrep satisfies $C(\rho) \geq C_F$, with equality exactly for F and $\bar{F}$ (the fundamental weights ω_1, ω_{N-1} are minuscule and $C(\lambda) = \langle \lambda, \lambda + 2\delta \rangle$ grows strictly along the dominance order). So a state at $2C_F$ has $k \leq 4$ charged links.

Geometry now does the rest, with no further representation arithmetic. A 1-valent site is never invariant, and a 2-valent site forces conjugate irreps on its two links, so the charged sublattice has minimum degree 2 at charged sites and any junction has valence ≥ 3. With $\sum_{\text{sites}} \deg = 2k \leq 8$: valence-3 junctions come in pairs ($3j + 2v = 2k$ with j even by parity of ends), and two 3-valent junctions with $k \leq 4$ require three disjoint paths of lengths summing to ≤ 4, hence lengths $(1, 1, 1)$ or $(1, 1, 2)$ — both demand two distinct links between one pair of adjacent sites, which the simple cubic torus does not have. A 4-valent junction with $k = 4$ requires all four links to close pairwise through 2-valent sites, again demanding parallel links. Cycles shorter than 4 need care, because the odd-L torus is *not* bipartite: exhaustive enumeration (machine-checked at $L = 3, 4, 5$) shows the only closed link sets of length < 4 are the $3L^2$ straight winding triangles at $L = 3$. A winding triangle carries one irrep ρ around a noncontractible cycle: if ρ has nonzero N-ality the trivial-centre-flux sector removes it, and if zero N-ality then $C(\rho) \geq C_{\text{Adj}} = N$ (the second classical input: the quasi-minuscule highest root minimises the Casimir on the nonzero root lattice), so its energy $3C(\rho)/2 \geq 3N/2 > 2C_F$. Two disjoint closed components need ≥ 8 charged links. So the charged sublattice at $E = 2C_F$ is a single 4-cycle of 2-valent sites: one irrep ρ runs around it (conjugated with orientation), the energy is $2C(\rho) = 2C_F$, and minimality forces $\rho \in \{F, \bar{F}\}$.

Finally, every 4-cycle of the torus at $L \geq 3$ is an elementary plaquette except, exactly at $L = 4$, the straight winding loops — machine-checked by exhaustive enumeration of all length-4 closed link paths at $L = 3, 4, 5$ — and the winding loops are removed by the sector qualification. The

E-eigenspace is therefore spanned by the two orientation states of each plaquette, and its charge-odd part by the $|p, -\rangle$. $\square$

Lemma 16 (Second-order process exhaustion)**.** *For $p \neq p'$ in the charge-odd one-plaquette sector, every nonvanishing second-order route $\langle p', -|VQ(E_{F,N} - H_E)^{-1}QV|p, -\rangle$ has its two V-insertions supported on the faces p and p' themselves. In particular, when p and p' share no link the off-diagonal element vanishes, and when they share one link it is exhausted by the four fusion channels of Theorem 11.*

Proof. V is a sum of single-plaquette terms, and multiplying by $\operatorname{Tr} U_q^{\pm 1}$ changes the electric content only on the four links of q; the resolvent is diagonal in the electric basis and changes none. Two insertions q_1, q_2 therefore leave every link outside $\operatorname{supp} q_1 \cup \operatorname{supp} q_2$ untouched, so $p \triangle p' \subseteq \operatorname{supp} q_1 \cup \operatorname{supp} q_2$. The set $p \setminus p'$ has three links (shared-link pair) or four (otherwise), and by the exhaustive shared-link geometry a face distinct from p contains at most one link of p: two insertions not supported on p cover at most two of them. Hence one insertion is supported on p, and symmetrically the other on p'; for $p \neq p'$ that pins $\{\operatorname{supp} q_1, \operatorname{supp} q_2\} = \{p, p'\}$ exactly.

It remains to dispose of the routes this still allows beyond the fusion channels. If p and p' are disjoint (sharing a vertex or nothing), both orders factor through a local charge parity: annihilating the p-flux costs $\langle 0_p|(\operatorname{Tr} U_p + \operatorname{Tr} U_p^\dagger)|p, -\rangle = 0$, the selection rule of Proposition 13, whether the intermediate is the vacuum or the two-plaquette state $|p\rangle \otimes |p'\rangle$; and the order that excites p first leaves adjoint flux on p that the remaining insertion, supported on p', cannot clear. If they share one link, the two insertion orders are precisely the two orientation families of Theorem 11, whose intermediate content on the shared link is the four fusion channels; an insertion orientation that drives a nonshared link into $\Lambda^2 F \oplus \operatorname{Sym}^2 F$ cannot be returned to the trivial representation by the other insertion and contributes zero. The diagonal $p = p'$ terms are unconstrained by this argument and remain the scalar of Corollary 12. $\square$

Theorem 17 (Global torus assembly, unconditional)**.** *For $N \geq 3$ and $L \geq 3$, in the sector of Lemma 15, the charge-odd effective Hamiltonian through second order is*

$$H^{(N)}_{\mathrm{eff},-}(k, u) = E_{\mathrm{flat},N}(u)I + t_N u^2 B(k)B(k)^\dagger + O(u^3),$$

with the three bands $E_{\mathrm{flat},N}(u)$ and $E_{\mathrm{flat},N}(u) + t_N u^2 q(k)$ twice, and flat carrier $\ker \partial_2$ of dimension $L^3 + 2$.

Proof. Lemma 15 discharges assumption (i) of Corollary 12 and Lemma 16 assumption (ii); the corollary's own proof then runs unchanged. $\square$

The status change deserves one sober sentence rather than celebration. What was conditional is now proved relative to the standard degenerate perturbation framework of (5), one classical Casimir-minimality fact, and finite combinatorics that a machine has enumerated; the charge-even identification of Section 6 remains a hypothesis, and nothing about convergence, higher orders, or the continuum has changed.

4.5 Finite-volume width

Let $q_{\max}(L)$ be the largest sampled q on the periodic L^3 grid. Then

$$q_{\max}(L) = \begin{cases} 12, & L \text{ even}, \\ 12\cos^2(\pi/2L), & L \text{ odd}, \end{cases} \tag{19}$$

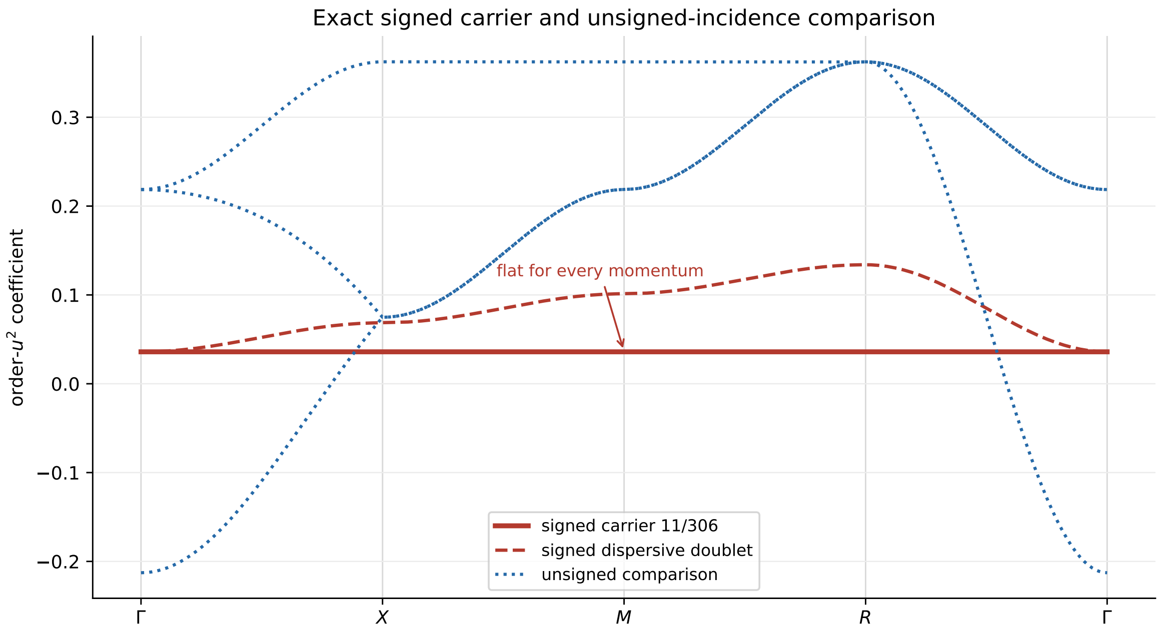

Figure 2: Order-u^2 coefficients along Γ–X–M–R–Γ. The red carrier and dispersive doublet are the signed incidence result; their conditional global assembly is Corollary 12. The blue curves are the exact unsigned-incidence comparison spectrum. Reading the blue curves as physical charge-even bands requires the separate symbol hypothesis of Section 6.

because only at even L does some k_j reach π exactly. The width of the full retained charge-odd manifold is $W^{(-)}_{N,L} = q_{\max}(L)\, t_N u^2 + O(u^3)$, so for even L, and coefficientwise as $L \to \infty$ at fixed perturbative order, $[u^2]W^{(-)} = 12t_N \sim 3/N^3$. This concerns the stiffness of the dispersive doublet, not motion of the lowest branch, which is flat at this order.

The smallest nonzero incidence eigenvalue is $q_{\min} = 4\sin^2(\pi/L)$, attained by one unit of momentum in a single direction.

4.6 The orientation signs are essential

Removing the orientation signs — from the entries of (6) and from the phases, so each $e^{ik} - 1$ becomes $e^{ik} + 1$ — gives an unsigned incidence $\mathcal{N}(k)$. Define its comparison adjacency by $A_+(k) := \mathcal{N}(k)\mathcal{N}(k)^\dagger - 4I$. Its adjacency spectra at Γ and R are $\{12, 0, 0\}$ and $\{-4, -4, -4\}$, which share no eigenvalue, so no level can be momentum independent. The flat branch is therefore not a generic consequence of three face orbitals or cubic symmetry.

Section 6 computes the unsigned comparison spectrum in closed form. The signed span is $8 - (-4) = 12$, giving the charge-odd second-order width $12t_3 = 5/51$ at $N = 3$. The unsigned continuous-zone span is $12 - (-4) = 16$. Interpreting $16|\ell_3| = 88/153$ as a charge-even bandwidth is conditional on deriving the complete linked charge-even symbol; on a finite torus the value -4 is sampled only when L is even.

5 The second-order coefficient on a two-cube space

Everything above derives t_N from a resolvent ledger indexed by the fusion channel of one shared link. That ledger is representation theory carried out on a single plaquette pair; it has never been tested against an operator calculation on a space large enough for two cubes to interact. This section reports such a test and, more importantly, states what it settles.

The construction is the open face-sharing $(3, 2, 2)$ prism with its eleven oriented plaquettes, in the exact-Casimir $\mathcal{B} = 6$ $SU(3)$ truncation, of basis dimension 1,590,462. The delivery reports that it reaches the relevant sector through 794 paths and 398 reachable states; it does not assemble a 1,590,462-dimensional magnetic matrix. Its complete charge-odd one-plaquette shell at $E_* = 8/3$ has dimension eleven — a census result and not the plaquette count. The full E_* eigenspace has dimension 22, split $11_{C=+} \oplus 11_{C=-}$ by tensor-derived charge conjugation, and the delivery removes all 22 from the pseudoinverse; because the eleven columns exhaust the odd half, that agrees with the $Q = 1_- - P_-$ prescription of (5). Writing K_{LR} for the raw second-order operator and K_L, K_R, K_F for the left-cube, right-cube and shared-face operators in their own coordinates, the connected operator is defined by the literal Möbius fold

$$K_{\text{conn}} = K_{LR} - J_L K_L J_L^\dagger - J_R K_R J_R^\dagger + J_F K_F J_F^\dagger, \tag{20}$$

and the same fold applied to the geometry gives G_{conn}. The fold is literal in the sense that no eigenvalue or fitted scalar is subtracted: each source operator is folded in its own coordinates and lifted. The delivery corroborates that internally by applying the same weights history by history over its 1,984 ordered two-step histories, of which 1,192 cancel at weight zero, and the two routes agree to 4.9×10^{-15}.

First order settles itself. On the shell $PVP = I_{11}$, so the first-order term is nonzero but scalar: it shifts all eleven branches equally and creates no branch ambiguity. Its connected fold then vanishes *identically*, and by counting rather than by cancellation — each of the eleven faces is covered exactly once by $\{L, R\}$ once the shared face is restored by F, so every Möbius weight in (20) applied to the identity is $1 - 1 = 0$. Second order is therefore the leading connected term and not a correction to a first-order one.

At second order the result is

$$K_{\text{conn}}^{(2)} = \frac{5}{612}\, G_{\text{conn}} + D, \tag{21}$$

with D diagonal and delivered. The geometry alone fixes the *splitting* inside each block and nothing else: G_{conn} turns out to be four disjoint 2×2 blocks plus three isolated directions (Appendix F), so each pair splits by $2 \times 5/612 = 5/306$ about its own diagonal entry. Given D, which Appendix F prints, the spectrum is then closed form: 0 once, $-\frac{15}{4}$ twice, $-\frac{34}{9}$ four times, and $-\frac{129}{34}$ four times — exact arithmetic on entries that are themselves reconstructions, a boundary made precise below. The delivery reports the coefficient recovered target-blind — supplied to no builder, graph fit, branching rule or validator — and reports that the payload hashes were produced before the recorded comparison, a chronology it itself calls record-backed and only partly auditable; the value is byte-present in the nested $\mathcal{B} = 4$ package that the $\mathcal{B} = 6$ build seals as an authority object, so the defensible statement is dependency-path nonuse and not corpus-wide absence. This paper re-thresholds those delivered gates rather than re-running the build, including a deliberately wrong-sign control that the held-out scaling rejects: on a 66-dimensional held-out word space the correct second-order residual falls with log-log slope 2.90 while the sign-reversed control falls with slope 2.00, a minimum error ratio of 325. What all this establishes is runtime nonuse of the comparator on the delivered dependency path, which is not a preregistration. The finite-u validations are asymmetric:

the $\mathcal{B} = 4$ delivery includes a held-out diagonalisation of the full 4,180-dimensional charge-odd space with an $O(u^3)$ residual, whereas the $\mathcal{B} = 6$ check is restricted to the 66-dimensional $P + Q_1$ star.

Resolving (21) by the irreducible representation the shared link actually carries in the ranked intermediate state gives six coefficients rather than four, because a link distinguishes ρ from $\bar{\rho}$:

$$\begin{array}{c|cccccc} \rho & \mathbf{1} & \mathbf{3} & \bar{\mathbf{3}} & \mathbf{6} & \bar{\mathbf{6}} & \mathbf{8} \\ \hline c_\rho & \frac{1}{12} & -\frac{1}{12} & -\frac{1}{12} & -\frac{1}{9} & -\frac{1}{9} & \frac{16}{51} \end{array} \tag{22}$$

That these six sum to 5/612 is weak evidence: six rationals reach one target in many ways, and a fitted set would pass it. The statement worth making is stronger, and it is what makes this more than a coincidence of one sum. First, though, the boundary.

The six coefficients of (22) are not symbolic: the delivery stores them as rational reconstructions of pinned finite-precision Clebsch–Gordan contractions, with a maximum reconstruction residual of 2.2×10^{-15} per channel — 2.1×10^{-14} across the delivery's reconstructed matrices as a whole — and its own certificate declaring `symbolic_exact_local_amplitudes: false`. What follows is therefore reported, conditional on those six numbers, and not a symbolic identity in the sense of Theorem 7.

Reported computational result 18 (The census is the Weingarten ledger)**.** At $N = 3$ the six reconstructed link-irrep coefficients of (22) are the four fusion-channel weights (14) individually:

$$c_{\mathbf{1}} = -w_{\mathbf{1}}, \qquad c_{\mathbf{8}} = -w_{\mathrm{Adj}},$$

$$c_{\mathbf{3}} = c_{\bar{\mathbf{3}}} = \tfrac{1}{2} w_{\Lambda^2 F}, \qquad c_{\mathbf{6}} = c_{\bar{\mathbf{6}}} = \tfrac{1}{2} w_{\mathrm{Sym}^2 F}.$$

Substituting (10) into (14) at $N = 3$ gives $w_{\mathbf{1}} = -1/12$, $w_{\mathrm{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$ and $w_{\mathrm{Sym}^2 F} = -2/9$; comparison with (22) is then arithmetic. Six reconstructed rationals landing on four predicted ones, with the two coincidences the correspondence requires, is a strong consistency statement and not a proof that the underlying contraction is exact.

One further feature of the census is worth separating out, because it is the paper's central architecture measured rather than assumed. The six link-resolved matrices are not merely six numbers that sum correctly: each is *separately* proportional to the same incidence Gram. The eleven oriented plaquettes of the prism have exactly 56 ordered adjacent pairs, each sharing one link with signed overlap ± 1; the delivery reports a constant channel-to-geometry ratio on all 56 of them, for every one of the six channels, with worst residual 2.2×10^{-15}. Colour dynamics fixes a number per channel and the cellular boundary operator fixes the matrix; if the two mixed, the ratio would be constant for the total and not for the parts.

The correspondence is scoped to what the Casimir closure exhausts: adjacent, one-action, shared-link intermediates at order u^2. Same-face diagonal routes, on-site terms and higher-action intermediates lie outside it — the $\mathcal{B} = 6$ scalar below is exactly a place where one of them is missing — and so does any untruncated-Hilbert-space statement.

It is scoped in one further way, which matters more. The two-cube build draws its local Wilson amplitudes from the same hash-pinned upstream Clebsch–Gordan archive as the single-cube reconstructions of the next subsection, and the squared amplitudes that archive produces are the published d_ρ/N^2 [26] — the numerator of (14). The census therefore does not confirm the weights from an independent source: it tests the *assembly* that carries them, which is the reachable-channel list, the resolvent denominators, the operator-level fold, the even split between conjugates, and the geometry that survives each channel separately. That is the statement worth making, and it is a different statement from confirmation.

Two features of the correspondence carry the content. The mixed family enters negated because $t_N = B_N - A_N$, which is the same relative sign the incidence orientation carries. And each like-family *fusion* weight is split evenly between the irrep and its conjugate — the two orientations of the shared link weighted equally — which the delivery resolves on the two-cube space rather than assuming. It is not measured *here*: what this repository checks is the identity the six delivered numbers satisfy.

Conjecture 19 (All-rank even split)**.** For every $N \geq 3$ each like-family fusion weight splits evenly between an irrep and its conjugate in the link-resolved census.

This is a conjecture and nothing here promotes it. Falsifying it means a link-resolved census at some $N \geq 4$ in a truncation that still retains every adjacent shared-link channel; at $N = 4$ the endpoint budgets make that $\mathcal{B} \geq 33/4$, and no such construction has been run.

5.1 One truncation, two constructions

The two-cube space also isolates why a cutoff can reverse the sign of t_3. An adjacent shared-link channel ρ is reachable only if the endpoint Casimir budget $C_2(\rho) + 2C_2(\mathbf{3})$ fits inside the truncation, meaning $C_2(\rho) + 2C_2(\mathbf{3}) \leq \mathcal{B}$: the sextet needs exactly 6 and the adjoint 17/3, so both survive $\mathcal{B} = 6$ and neither survives $\mathcal{B} = 4$. The weak inequality is not a convenience. Both retentions this section uses are decided by an *equality*: $\Lambda^2 F = \bar{\mathbf{3}}$ has budget exactly 4 and so sits exactly on the $\mathcal{B} = 4$ cutoff, and the sextet sits exactly on the $\mathcal{B} = 6$ one. Under a strict rule $\mathcal{B} = 4$ would reach $\{\mathbf{1}\}$ alone and $\mathcal{B} = 6$ would lose the sextet, contradicting both delivered censuses — so the deliveries are what fix the convention, which is also why the budget arithmetic is not an independent confirmation of either census.

Corollary 20 (What a link cutoff drops)**.** *Retaining only the singlet and $\Lambda^2 F$ routes projects the second-order hopping to*

$$w_{\Lambda^2 F} - w_{\mathbf{1}} = -\tfrac{1}{12},$$

opposite in sign to t_3 and 10.2 *times larger in magnitude. What is dropped is*

$$w_{\mathrm{Sym}^2 F} - w_{\mathrm{Adj}} = \tfrac{14}{153},$$

and $-\frac{1}{12} + \frac{14}{153} = \frac{5}{612} = t_3$. This is arithmetic on (14) *alone.*

The published $p + q \leq 1$ link cutoff retains exactly those two routes — a per-link restriction, where $\mathcal{B} = 4$ is a total endpoint budget, so the two schemes agree on the retained channels at this order and are not the same rule — so $-1/12$ is its projected adjacent hopping and 14/153 is the adjacent-hopping correction that completes it — not 5/612, which would double-count what it already carries. On the two-cube space the first of those two numbers is delivered rather than inferred: a separately sealed $\mathcal{B} = 4$ run on the identical geometry gives $K^{(2),\mathcal{B}=4}_{\mathrm{conn}} = -\frac{1}{12} G_{\mathrm{conn}} + D_{\mathcal{B}=4}$ outright, and the Casimir budgets of the previous paragraph say without reference to any census that $\mathcal{B} = 4$ can reach only $\{\mathbf{1}, \mathbf{3}, \bar{\mathbf{3}}\}$. Conditional on (22), the same split of the census reproduces it, $c_{\mathbf{1}} + c_{\mathbf{3}} + c_{\bar{\mathbf{3}}} = -1/12$, and gives the omitted half as $c_{\mathbf{6}} + c_{\bar{\mathbf{6}}} + c_{\mathbf{8}} = 14/153$.

Remark 21. Two single-cube reconstructions bracket the same statement. A full $\mathcal{B} = 4$ open cube assembled from the pinned `ymcirc` $d = 3$, $\mathcal{B} = 4$ plaquette data [27, 28] reproduces $-1/12$ and the truncated shell ordering to 4.3×10^{-11}; a $\mathcal{B} = 6$ cube reproduces $+5/612$ and the inverse ordering to 8.4×10^{-15}. Both locate the missing weight at the same 14/153.

Neither is an independent replication. Both read tabulated plaquette matrix elements of the same lineage — and the two-cube build draws its Clebsch–Gordan amplitudes from the same hash-pinned `pyclebsch` archive as the $\mathcal{B} = 6$ cube, so that agreement is a cross-check of assembly and not of the amplitudes. The published closed form [26] is d_ρ/N^2 — the very numerator of (14). This is a cross-check of assembly, not a third confirmation of t_3; the exact rational identity is Corollary 20, which needs no tolerance. The $p + q \leq 1$ nomenclature and the finite-rank matrix element are both from [26], and single-cube $SU(N)$ diagonalisation in this style goes back to [16].

The correction to the adjacent-hopping coefficient of a published T_1 Hamiltonian is therefore 14/153 and not 5/612, which would double-count the two routes T_1 already carries. One coincidence of value is worth naming: $-1/12$ is also the singlet weight $w_{\mathbf{1}}$, and the cutoff value is a *difference* of two weights, not a weight.

The $\mathcal{B} = 6$ cube is channel-complete for the hopping and *not* for the scalar: its scalar is 39/68 against the bridge's 11/34, differing by exactly 1/4, the local same-face sextet route, which first enters at cutoff 20/3 — integer label $\mathcal{B} = 7$, so any cutoff above 20/3 already carries it.

5.2 What the cutoff moves, and where

The connected diagonal D is eleven numbers in both deliveries and is treated as a remainder in both. It is not eleven numbers. The open prism has a symmetry group — the cell exchange $x \mapsto 2 - x$, the two transverse reflections and the $y \leftrightarrow z$ swap, of order 16 — and D is constant on its orbits, which are $8 + 2 + 1$: the eight side faces, the two end caps, the shared face.

Proposition 22 (Where the truncation lives)**.** *The eight-element orbit is exactly the support of the four cross-cell pairs, and it is the only orbit whose diagonal value differs between the two truncations: the end caps carry* $-15/4$ *and the shared face* 0 *at both* $\mathcal{B} = 4$ *and* $\mathcal{B} = 6$*, while the side value moves from* $-7/4$ *to* $-2317/612$*. Together with the four cross-cell blocks carrying the whole off-diagonal change, the entire* $\mathcal{B} = 4 \to \mathcal{B} = 6$ *difference is confined to the faces that carry connected transport.*

This also says what the nonuniform D is. Where the faces of a complex form a *single* orbit — the closed cube of the next subsection, or the periodic torus under Corollary 12 — orbit-constancy forces the diagonal to be scalar, and the operator is exactly $\alpha I + tG$. The prism has three orbits because it has a boundary. D is the open boundary showing up in the diagonal, not a failure of the incidence form.

5.3 The one-cube shell, and why its flat level cannot move

Those two single-cube reconstructions are usually read as numerical cross-checks. They are more than that, because the object they diagonalise is the same chain complex. The six faces of a cube are a closed surface, so in the outward orientation $\ker \partial_2$ is one-dimensional and spanned by the sum of all six — the fundamental class, $b_2(S^2) = 1$. That is Theorem 2 on a complex with no three-cells, where the $L^3 - 1$ boundaries are absent and only the class survives.

Proposition 23 (One-cube shell)**.** *On the closed cube surface* $G = \partial_2^\top \partial_2$ *commutes with all* 24 *proper rotations. The combinations* $b_{+i} - b_{-i}$ *carry the defining vector representation and the combinations* $b_{+i} + b_{-i}$ *the permutation representation of the three unordered axes, so with the charge-odd inversion* $Pb_{+i} = -b_{-i}$ *the shell is*

$$A_1^{--} \oplus T_1^{+-} \oplus E^{--},$$

with G*-eigenvalues* 0*,* 4*,* 6 *and multiplicities* 1*,* 3*,* 2*. Consequently* $\alpha I + tG$ *has levels* α*,* $\alpha + 4t$*,* $\alpha + 6t$*.*

Proof. The rotations permute the six outward faces, and G is built from the boundary map, so it commutes with the permutation action. On the antisymmetric combinations that action is the defining representation by inspection — $b_{+i} - b_{-i}$ transforms as e_i — and on the symmetric combinations it is the permutation representation of $\{\pm e_i\}$, which is $A_1 \oplus E$. Under $Pb_{+i} = -b_{-i}$ the antisymmetric sector is parity even and the symmetric one parity odd. In these sectors G is $4I$ and $4I - 2(J - I)$ respectively, of spectra $\{4, 4, 4\}$ and $\{0, 6, 6\}$; since the three isotypic components have distinct dimensions the eigenvalues attach to them uniquely. □

Three things follow. The cubic labels are derived here rather than assigned: both deliveries assert them and neither builds the symmetry matrices. The A_1^{--} level is Proposition 33's mechanism on a second complex — $Gw = 0$ there, so that level sits at α for every t, and no truncation, no channel and no coefficient in dispute anywhere in this paper can move it. And the reversal the two runs report is therefore a statement about the sign of one number rather than about the shell as a whole: at $\mathcal{B} = 6$ the levels are {39/68, 371/612, 127/204} and at $\mathcal{B} = 4$ {37/12, 11/4, 31/12}, in the opposite order, with A_1^{--} in each case sitting exactly at the scalar. The relative shell $\{0, 4t, 6t\}$ is {0, 5/153, 5/102} at $\mathcal{B} = 6$.

The finite-volume fingerprint of the 28 August finite-N bridge certificate — a different release, and named here because a reader tracing it to the two-cube one would find nothing — is worth recording because its most discriminating multiplicity is not a feature of the truncation: on the periodic 3^3 torus the incidence eigenvalues $0, 3, 6, 9$ have multiplicities $29, 12, 24, 16$, respectively, and $29 = L^3 + 2$ is Theorem 2.

Remark 24 (Scope). Everything in this section is second order in u, finite volume, strong coupling, one sector. It bears on t_N and decides nothing about the fourth-order coefficient of Section 9. The $\mathcal{B} = 6$ two-cube construction was not re-run here: its arithmetic is audited against geometry rebuilt independently, and its diagonal D is B-truncated and not proved cutoff-stable. The held-out finite-u validation that rejects the wrong-sign control is a 66-dimensional $P + Q_1$ word space with the Q_1–Q_1 magnetic block set to zero, not the full Hamiltonian, and no radius-three stability claim was executed. The two branches are not equally covered: at $\mathcal{B} = 4$ the delivery reports a direct diagonalisation of the complete 4,180-dimensional charge-odd block at four small couplings, two of them held out, with the residual after $E_* + u + u^2 \operatorname{eig} K_2$ scaling as u^3; at $\mathcal{B} = 6$ no such diagonalisation exists and the 66-dimensional star stands in for it. That $\mathcal{B} = 4$ report carries no check here — its certificate did not travel — and is recorded as the delivery's rather than as this paper's. No external group has reproduced the release.

5.4 A cutoff-free radius-two witness at finite coupling

A later sealed delivery (29 August 2026) assembles the charge-odd and charge-even $SU(3)$ two-cube Ritz operators with no Casimir cutoff at all: the electric seed P (eleven charge-odd faces), the complete one-action frontier Q_1, and the exact-energy-factorized two-action frontier Q_2, with cumulative odd dimensions 11, 77, 773. Three nested operators are diagonalised at each coupling — K_1 on $P + Q_1$; K_2^*, which zeroes exactly the Q_2–V–Q_2 block; and the full K_2 — and every finite-coupling Cauchy interlacing inequality between them is checked, with recorded violation exactly 0. All quoted quantities are invariant under diagonal basis rephasing, which is why no signed hopping entry is read as an observable there. The delivery is finite-precision throughout; everything in this subsection is reported at stated tolerances, never promoted.

Reported computational result 25 (The rational sub-block of the cutoff-free second-order spectrum)**.** Six of the eleven eigenvalues of the sealed cutoff-free second-order block $C_2 = BR_1B^\dagger$

on the odd seed are rationals to within 10^{-12} (the worst residual is 5×10^{-14}, about 30 ulps):

$$-\tfrac{2429}{306}\ (\times 2), \qquad -\tfrac{404}{51}\ (\times 3), \qquad -\tfrac{2419}{306}\ (\times 1),$$

and the outer levels straddle the central triple by exactly

$$\tfrac{2429}{306} - \tfrac{404}{51} = \tfrac{404}{51} - \tfrac{2419}{306} = \tfrac{5}{306} = 2t_3.$$

The remaining five eigenvalues are not rational at this precision.

The point is what this operator was *not* told. The census of Reported result 18 recovered the shared-link channel coefficients from a build that retained those channels; here the sealed radius-two operator retains every second-order route the two-action frontier supports, makes no shared-link truncation, and the shared-link coefficient $t_3 = 5/612$ of Theorem 11 appears anyway — as the exact splitting scale of a rational sub-block, embedded in a spectrum whose other levels are algebraic. A structural identification of that sub-block (which face combinations carry it, and why the central level is threefold) is not derived here and is recorded as a lead, not a claim; the open two-cube cluster has face-inequivalent environments, so a non-scalar diagonal is expected and no contradiction with Corollary 12, which is a torus statement, arises.

The same delivery reports direct nested-difference coefficients at fourth and fifth order — for the matched odd–even gap, $C_4^{\mathrm{gap}} = -0.8272230$ and $C_5^{\mathrm{gap}} = +4.1838863$ — each agreeing with an independent weak-coupling Hellmann–Feynman intercept to better than 3×10^{-6}. These are increments between the nested truncations, in the delivery's own coupling convention, and are not asserted to be complete fourth- or fifth-order coefficients of the unablated sector Hamiltonian; no convergence claim accompanies them.

One further feature bears on Section 9. The delivery's fifth-order object on the eleven-dimensional seed is recorded as a *matrix*: its traceless part is nonzero, with $\mathrm{tr}(\mathrm{shape}^2) = 0.359$ and eleven recorded shape eigenvalues. It is therefore the first delivered instrument in this corpus whose output is not a Γ-point scalar. That observation is about the instrument class only: the two-cube cluster carries no off-axis momentum, and nothing here weakens Theorem 35 or bears on the value of C_{shp}.

6 The unsigned-incidence comparison operator

The unsigned incidence $\mathcal{N}(k)$ of Section 4 defines an exact algebraic comparison operator $G_+(k) = \mathcal{N}(k)\mathcal{N}(k)^\dagger$. Its identification with the complete linked charge-even effective Hamiltonian is an additional hypothesis, not a result of the shared-link calculation: Proposition 13 exposes an all-pairs vacuum route whose cancellation or linked-cluster reorganisation has not been derived. With that boundary explicit, write

$$\hat{a}_m = 2 + 2\cos k_m = |1 + e^{ik_m}|^2, \qquad p = \sum_m \hat{a}_m.$$

The hat is not decoration: $\hat{a}_m$ is the *complement* of the $a_m = 4\sin^2(k_m/2)$ used in the charge-odd shape basis (25) below, $\hat{a}_m + a_m = 4$, and that complementarity is the sum rule below.

Theorem 26 (Unsigned-incidence Bloch cubic)**.** *$p + q = 12$ pointwise, and the characteristic polynomial of $\mathcal{N}\mathcal{N}^\dagger$ is*

$$\mu^3 - 2p\,\mu^2 + p^2\mu - 4\hat{a}_1\hat{a}_2\hat{a}_3 \;=\; \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3,$$

with the unsigned comparison-adjacency eigenvalues $\lambda = \mu - 4$. The signed counterpart of the same computation is $\mu(\mu - q)^2$.

The algebraic closed form is complete rather than merely consistent at sampled momenta: the characteristic polynomial of the 81×81 and 192×192 finite plaquette adjacencies factorises coefficient for coefficient over $\mathbb{Z}$ into this cubic over the momentum grid.

Proposition 27 (Range and attainment)**.** *$\lambda \in [-4, 12]$ exactly. The top $\lambda = 12$ is attained only at Γ ($\hat{a}_1 = \hat{a}_2 = \hat{a}_3 = 4$). The bottom $\lambda = -4$ is attained on the whole of the three planes $k_j = \pi$, where the constant term vanishes, and only at R is it a triple root.*

Proof. Write $f(\mu) = \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3$. Then $f(0) = -4\hat{a}_1\hat{a}_2\hat{a}_3 \le 0$ and $\mu(\mu - p)^2 < 0$ for $\mu < 0$, so no root lies below 0; and $f'(\mu) = (3\mu - p)(\mu - p) > 0$ for $\mu > p$, with $\hat{a}_1\hat{a}_2\hat{a}_3 \le (p/3)^3$ by AM–GM and

$$f(16) \ge 16(16 - p)^2 - \frac{4p^3}{27} = \frac{4}{27}(12 - p)(48 - p)^2 \ge 0.$$

Hence $\mu \in [0, 16]$ and $\lambda = \mu - 4 \in [-4, 12]$. Equality at the top forces $p = 12$ and $\hat{a}_1\hat{a}_2\hat{a}_3 = 64$, which is Γ. Equality at the bottom is $f(0) = 0$, i.e. $\hat{a}_1\hat{a}_2\hat{a}_3 = 0$, i.e. some $k_j = \pi$; there $f(\mu) = \mu(\mu - p)^2$ and the root $\mu = 0$ is simple unless $p = 0$, which is R alone. ☐

Remark 28. A flat unsigned-comparison branch would need $f(\mu_0) = 0$ identically in k, which the constant term already forbids. Its minimum is attained on three measure-zero planes, against the signed operator's flat band. This contrast is an exact statement about two incidence constructions, not by itself a derivation of two physical charge sectors.

6.1 Conditional charge-even assembly

The shared-link coefficient at $r = 2$ is all-rank and analytic. Turning it into a charge-even band structure assumes the symbol identification stated above; the displayed second-order scalar additionally uses the supplied $SU(3)$ leakage/tower identification. The $r = 3$ values — $t_{3,+} = -6335/249696$ and everything computed from it — come from the same supplied domino ledger as Section 7, and are conditional on it in the same way.

Conditional on that identification, the scalar ledger is not a list of separate facts. Writing λ for the adjacency eigenvalue and $\text{tower}_{r,s}$ for the within-plaquette term in canonical u,

$$E_s(\lambda, r) = \text{tower}_{r,s} + 12\,\text{leak}_{r,s} + \lambda\, t_{r,s}, \tag{23}$$

with $\lambda \in \{-4, 8\}$ in the signed charge-odd sector (carrier and dispersive top), $\lambda \in \{12, -4\}$ in the unsigned comparison, and $\lambda = 0$ returning the on-site diagonals. This reproduces every registered diagonal and band value — nine of them — across both sectors and both orders. The tower term is the certified coupling conversion $4\Delta_\pm(3u/2)$, where at $N = 3$ the within-plaquette one-plaquette series of the two sectors are

$$\begin{aligned} \Delta_-(x) &= \tfrac{2}{3} + \tfrac{x}{6} + \tfrac{1}{18}x^2 + \tfrac{7}{432}x^3, \\ \Delta_+(x) &= \tfrac{2}{3} - \tfrac{x}{6} + \tfrac{13}{180}x^2 + \tfrac{101}{2700}x^3, \end{aligned} \qquad x = \frac{\beta_3}{4} = \frac{3}{2}\,u, \tag{24}$$

in the corpus's own printed coefficients; so (23) takes no unregistered input. The letter Δ is reused elsewhere with other subscripts — the fourth-order checkpoint differences, the coefficient gap Δ_C — and only the sector subscript $\pm$ means this series. The twelve is a count on this lattice, not a citation: the plaquette graph is 12-regular and two distinct faces share at most one link. The two towers are different objects and both are needed: the charge-odd one contributes $8/3 + u + u^2/2 + 7u^3/32$ and the charge-even one $8/3 - u + (13/20)u^2 + (101/200)u^3$, each of them $4\Delta_\pm(3u/2)$ on its own sector's series, so $\text{tower}_{2,-} = 1/2$ against $\text{tower}_{2,+} = 13/20$ and $\text{tower}_{3,-} = 7/32$ against $\text{tower}_{3,+} = 101/200$.

Because the supplied ledger has $\mathrm{leak}_{r,+} = t_{r,+}$ at both computed orders, the conditional charge-even assembly collapses to $\mathrm{tower}_{r,+} + (12+\lambda)\, t_{r,+}$, a one-parameter family in the charge-even tower. That equality is verified and unexplained. The order-three separation is between the charge-odd leakage and the charge-*even* hopping — not, as an earlier edition printed, between the charge-odd leakage and the charge-odd hopping, which were never equal: $\mathrm{leak}_{2,-} = t_{2,+} = -11/306$ against $t_{2,-} = 5/612$ at second order, and $\mathrm{leak}_{3,-} = -12331/249696$ against $t_{3,+} = -6335/249696$ at third. It is recorded as a unifying candidate with a falsifier — an order at which the two are computed independently and differ — not as a mechanism.

6.2 Where the rest frame is not blind

Conditional corollary 29 (Charge-even Γ splitting)**.** If the unsigned symbol is the complete linked charge-even symbol, then at Γ its comparison spectrum is $\{12, 0, 0\}$, so the A_1^{++} level and the E^{++} doublet are distinct, and

$$E(A_1^{++}) - E(E^{++}) = 12\, t_{r,+}$$

exactly: $-22/51$ at $r = 2$ and $-6335/20808$ at $r = 3$.

This is the structural counterpart of the charge-odd blindness recorded in Section 7. The signed spectrum at Γ is $\{-4, -4, -4\}$, one level, so no charge-odd rest-frame series can see the hopping; the unsigned comparison spectrum has two levels, so the charge-even rest frame could fix the coefficient under the symbol hypothesis. Neither half needs the incidence ansatz, and stating them through it understates both. At Γ the little group is the full cubic point group, so the 3×3 block commutes with its action at every order in u and whether or not the correction factors through links; Schur then decides the two sectors, and decides them differently. The charge-odd state is the *oriented* 2-cell $e_i \wedge e_j$, on which a proper rotation acts by its exterior square, which is the rotation itself under Hodge duality — the defining rep T_1, irreducible, commutant dimension one, so the block is a scalar. Charge conjugation drops the orientation, so the comparison state is the *unoriented* plane and the action is the permutation of the three axes, $A_1 \oplus E$, commutant dimension two. The two commutant dimensions are exactly the two level counts, and this half of the statement carries no symbol hypothesis at all.

The A_1^{++} branch is isotropic at leading order without assuming it, with curvature $\frac{4}{3}|t_{r,+}|$ in all three directions — 22/459 at second order and 6335/187272 at third.

On a finite L^3 grid the conditional charge-even width is

$$W_{r,L}^{(+)} = |t_{r,+}|\, [12 - \lambda_{\min}^{(+)}(L)], \qquad \lambda_{\min}^{(+)}(L) = -4 \iff L \text{ is even.}$$

Thus it equals $16|t_{r,+}|$ for even L and approaches that coefficientwise as $L \to \infty$; the charge-odd manifold width is $12|t_{r,-}|$ for even L. At any order at which the correction stays in the incidence algebra — which Proposition 33 warns is not automatic — the second-order ratio is $4(3N^2 - 5)/[3(N^2 - 4)]$, finite and greater than one at every rank $N \geq 3$, decreasing from 88/15 at $N = 3$ towards 4. So at second order the charge-even band is the wider one at every rank under the charge-even symbol hypothesis, and that asymmetry is a channel statement rather than a geometric one.

7 $SU(3)$ through third order

The *third-order* content of this section is an externally verified computational input: the frozen upstream corpus reports a 251/251 cold exact replay for the retained chain, while this compact release checks only the assembled rationals. At second order the hopping coefficient t_3 is analytic, but the displayed flat scalar additionally uses the upstream $SU(3)$ identification $\mathrm{leak}_2 = \ell_3$ and its within-plaquette tower. That tower contributes $8/3 + u + u^2/2 + 7u^3/32$.

Reported computational result 30 (Third-order ledger)**.** The supplied ledgers report, in exact rationals,

$$b_3 = \frac{1975}{124848}, \qquad \mathrm{leak}_3 = -\frac{12331}{249696},$$

and the assembled flat scalar $d_3 = \frac{7}{32} + 12\,\mathrm{leak}_3 - 4b_3 = -\frac{109151}{249696}$.

Conditional corollary 31 (Third-order factorisation)**.** If the lifter census is exhaustive and the linked ledger complete, then

$$H_{\mathrm{eff},-}(k,u) = E_{\mathrm{flat}}(u)I + \tau(u)B(k)B(k)^\dagger + O(u^4),$$

$$E_{\mathrm{flat}}(u) = \tfrac{8}{3} + u + \tfrac{11}{306}u^2 - \tfrac{109151}{249696}u^3,$$
$$\tau(u) = \tfrac{5}{612}u^2 + \tfrac{1975}{124848}u^3.$$

The second-order diagonal is the $\lambda = 0$ value $\frac{1}{2} + 12\,\mathrm{leak}_2 = 7/102$ with $\mathrm{leak}_2 = -11/306$; subtracting $4t_3$ gives the flat scalar $11/306$ at $\lambda = -4$.

Remark 32 (What the external comparison can and cannot do). Hamer's 1989 rest-frame series [13] agrees with E_{flat} at every shared order. It cannot test τ. The hopping enters the spectrum only through $\tau(u)q(k)$ and $q(0) = 0$, so every *charge-odd* Γ-point datum is blind to it. Under the unsigned-symbol hypothesis the charge-even rest frame is not, by Conditional result 29; the one-parameter family $(b_3 + \varepsilon,\ \mathrm{leak}_3 + \varepsilon/3)$ leaves d_3 and therefore the entire comparison unchanged. What the agreement pins is the combination $12\,\mathrm{leak}_3 - 4b_3$. The $\lambda = 8$ band top separates them, via (23). Conditional result 29 is the charge-*even* analogue and separates leak_+ from t_+; it says nothing about b_3 or leak_3.

8 What the homology protects

Proposition 33 (Boundary-factorised rigidity)**.** *If an order-r Bloch correction has the form $K_-^{(r)}(k) = a_r I + b_r S(k) + B(k)M_r(k)B(k)^\dagger$, then for every $k \neq 0$, $K_-^{(r)}(k)w(k) = (a_r - 4b_r)w(k)$: the correction shifts the carrier without dispersing it.*

Proof. $B^\dagger w = 0$ and $Sw = -4w$, both from Theorem 1. The middle term is annihilated whatever M_r is, which is the content: the protection belongs to the boundary operator and not to the dynamics. ☐

In real space this is the orthogonality of the two cellular Laplacians, $L_2^\downarrow L_2^\uparrow = L_2^\uparrow L_2^\downarrow = 0$, so any polynomial in them acts on $\ker L_2^\downarrow \cap \ker L_2^\uparrow$ by its constant term. The proposition is not an unconditional all-orders statement: $\{BMB^\dagger\}$ is not a two-sided ideal in $\mathrm{End}(C_2)$, and a general higher-order correction need not factor through links. The carrier-projected fourth-order shapes the corpus works with are

$$1, \quad q, \quad e_2, \quad \frac{4e_2}{q}, \quad \frac{e_3}{q}, \tag{25}$$

with $e_2 = \sum_{i<j} a_i a_j$, $e_3 = a_1 a_2 a_3$ and $a_i = 4\sin^2(k_i/2)$. The $1/q$ is the carrier projection: $\|w\|^2 = q$ by Theorem 1, so the carrier-projected correction is $\varepsilon_4 = (w^\dagger K_-^{(4)} w)/q$ and the numerator, not ε_4, is the polynomial.

That family is *not* what cubic symmetry permits, and an earlier edition attributed it there. The point group permutes (a_1, a_2, a_3), so its invariants are the symmetric polynomials, and the numerators of degree at most three with no constant term span a six-dimensional space $\{q, q^2, e_2, q^3, qe_2, e_3\}$ — one per partition. Clearing the denominator in (25) returns five of those six; the missing direction is q^3, that is the shape q^2, which is cubic invariant and regular at Γ ($q^2 \sim |k|^4$), so neither the point group nor the carrier projection excludes it. The omission is a property of the two-hop enumeration that produced the family, and it is load-bearing: Theorem 35 is a rank statement about *this* span, and a sixth direction would be a sixth unknown for the same data to determine.

9 The fourth-order boundary

Two supplied fourth-order records agree on the axial datum and on the appearance of mobility, and disagree on one off-axis mixed-gradient coefficient $C_{\rm shp}$. This edition does not adjudicate them, and no fourth-order band or bandwidth is used above. What it does is state precisely where the disagreement lives, because that geography is now checked and it is what a decisive calculation must target.

After subtracting the rest value, the recorded carrier ansatz is

$$\Delta_4(k) := \varepsilon_4(k) - \varepsilon_4(0) = Aq + Be_2 + C_{\rm shp}\frac{4e_2}{q} + D\frac{e_3}{q}. \tag{26}$$

The continuous $k \to 0$ limit is used at $q = 0$. Denote the two recorded mixed-gradient values by $C_{\rm old}$ and $C_{\rm new}$ and set $\Delta_C = C_{\rm new} - C_{\rm old}$. A supplied all-rank cube ledger also reports

$$\alpha_N = \frac{640}{N(N^2-1)^3}, \tag{27}$$

but this formula and its identification with the complete axial coefficient remain conditional on that ledger. Sealed packages containing other fourth-order scalars use incompatible observable labels; none is promoted here to a canonical rest coefficient.

9.1 An exact theorem for the saved historical kernel

The upstream authority corpus supplies the hash-pinned 189-record historical kernel. Pulling its centered fourth-order operator back to cube amplitudes with $\mathsf{C} = \partial_3$, define

$$Q_4 = \mathsf{C}^\dagger K_-^{(4)} \mathsf{C}, \qquad G = \mathsf{C}^\dagger \mathsf{C}, \qquad \mathcal{Q}_4 = Q_4 - s_4 G,$$

where $s_4 = \operatorname{tr} K_-^{(4)}(\Gamma)/3$. If $L_i = 2I - T_i - T_i^{-1}$, then $G = \sum_i L_i$ and the centered coefficient is the generalized eigenvalue $\mathcal{Q}_4\phi = \lambda_4 G\phi$. The representative change $(Q_4, G) \sim (Q_4 + \delta G, G)$ is a scalar gauge only when the scalar anchor changes simultaneously; it cannot change an off-axis shape.

Reported computational result 34 (Exact saved-kernel Hodge pencil)**.** For the supplied historical kernel,

$$q^{(4)}_{\rm old} = -\frac{20721577909065127111}{7250590288602460800}, \qquad \alpha_{\rm old} = \frac{5}{12}, \qquad \beta_{\rm old} = \frac{17607806155349}{275331901291200},$$

and

$$\mathcal{Q}_{4,\rm old} = \frac{5}{48}\sum_i L_i^2 + \frac{17607806155349}{1101327605164800}\sum_{i<j} L_i L_j \succeq 0. \tag{28}$$

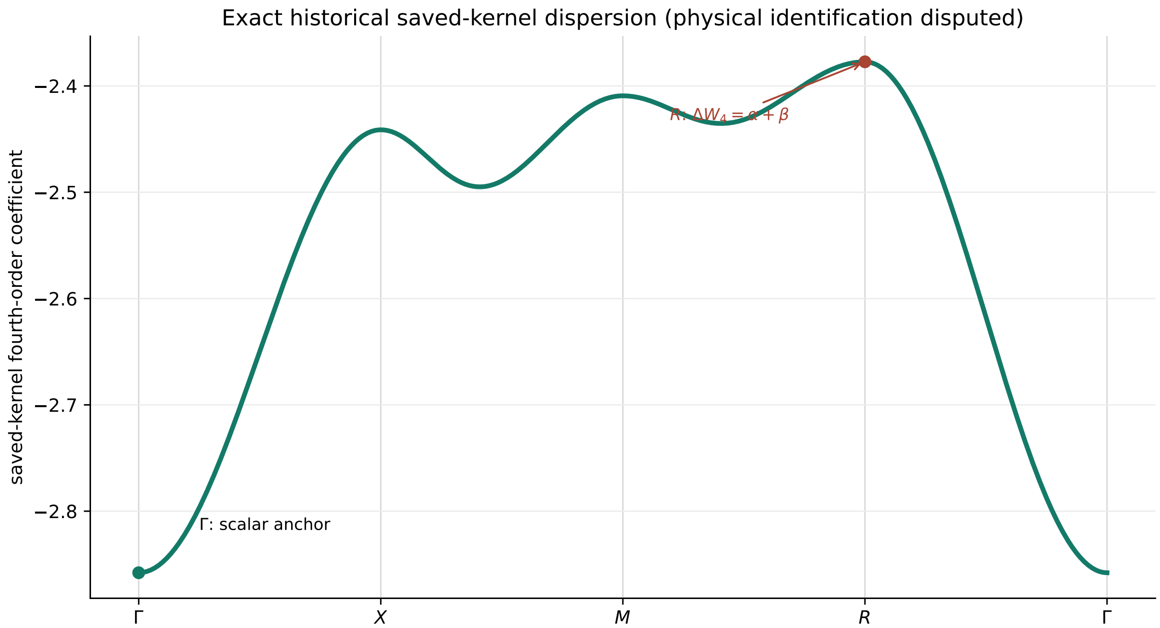

Figure 3: Exact dispersion of the saved historical kernel in Reported result 34. The plot is a theorem about that hash-pinned kernel. Its identification with the complete physical linked $SU(3)$ fourth-order operator is disputed, so the curve is not used as a settled physical band.

Its continuous-zone and even-L width is

$$W_{4,\text{old}} = \alpha_{\text{old}} + \beta_{\text{old}} = \frac{132329431693349}{275331901291200}.$$

These identities are exact for the saved kernel and its cold fixed-kernel checks. They do not establish that this kernel is the unique physical linked fourth-order operator.

The corresponding 25-point numerator stencil has

$$w_0 = \tfrac{9}{2}\alpha + 3\beta, \qquad w_1 = -(\alpha+\beta), \qquad w_2 = \tfrac{1}{4}\alpha, \qquad w_d = \tfrac{1}{4}\beta,$$

and obeys $w_0 + 6w_1 + 6w_2 + 12w_d = 0$ exactly. For $\alpha, \beta > 0$, the generalized dispersion is bounded by $0 \le \lambda_4 \le \alpha + \beta$, with the continuous-zone extrema at Γ and R respectively; an odd-L grid does not contain R.

The two records do agree on the axial datum, but the corpus's own quoted tolerance does not cover that agreement: the α gap is 2.43×10^{-13} against a stated bound of 2.3×10^{-13}. The sealed core holds internally — α_{new} and $4A_{\text{new}}$ agree to 2×10^{-15} — which is consistent with the quoted bound being too tight, but two float comparisons do not discriminate that from a shared upstream slip. Recorded as a discrepancy rather than resolved, and the bound is not widened.

9.2 The obstruction is polynomial

Theorem 35 (Obstruction certificate)**.** *The two recorded off-axis coefficients enter* (25) *only through the shape* $4e_2/q$*, so the two records' carrier numerators differ by* $4\Delta_C e_2$ *and their bands by* $\Delta_C\,(4e_2/q)$*. On any one-dimensional axial cut two of the three* a_i *vanish, so* e_2 *is the* zero polynomial *there and both differences vanish identically. Consequently no* Γ*-point or axial datum* *distinguishes the two records, at any precision. Off axis they do differ:* $e_2(M) = 16$ *and* $e_2(R) = 48$*, so the numerators differ by* $64\Delta_C$ *and* $192\Delta_C$ *and the bands by* $8\Delta_C$ *and* $16\Delta_C$*.*

Remark 36. The vanishing is polynomial, not numerical: this is non-identifiability, not a limitation of the data in hand. Re-anchoring cannot substitute, and neither record is preferred here. An off-axis contraction is the only decider.

On the axial cut itself the invariants collapse to $q = q_{\mathrm{ax}}(k) = 4\sin^2(k/2)$ with $e_2 = e_3 = 0$. Dividing the raw cube-boundary numerator $(\alpha_N/4)q_{\mathrm{ax}}^2$ by $\|w\|^2 = q = q_{\mathrm{ax}}$ leaves one power of q_{ax} with coefficient $\alpha_3/4 = 5/48$.

9.3 A second witness

Non-identifiability is better exhibited than argued from a difference between two records, since a difference could always be a computational error in one of them. The exhibition adds no logical strength: Theorem 35 already proves that *every* value of C_{shp} is consistent with the retained data, so a witness is one point of a line already known to be a line. It is a concrete handle, not evidence. The witness below is not itself a candidate for the physical coefficient. It is the balanced continuation obtained in the supplied source by setting an auxiliary bookkeeping parameter ϵ to zero; that source label carries no physical meaning here. Its sole role is to make the $4\Delta_C e_2$ family concrete with an exactly known shift.

Proposition 37 (Explicit second witness)**.** *There is a value* C_{alt} *with* $C_{\mathrm{alt}} - C_{\mathrm{old}} = 25/1024$ *exactly, appearing verbatim in the pinned source edition, whose induced band separations are* 0 *at* X*,* $25/128$ *at* M *and* $25/64$ *at* R*. It is identical to* C_{old} *on every retained datum and distinct from it off-axis.*

9.4 Where the coefficient lives

The four checkpoint deltas are the Γ-subtracted carrier corrections $\Delta_s = \varepsilon_4(k_s) - \varepsilon_4(0)$ at $k_X = (\pi, 0, 0)$, $k_M = (\pi, \pi, 0)$, $k_P = (\pi, \pi/2, 0)$ and $k_R = (\pi, \pi, \pi)$ — P is the off-axis, off-corner point that makes the system invertible, with $a = (4, 2, 0)$. Its algebraic role is explicit here; no additional provenance is inferred for that checkpoint. In the shape basis they are

$$\begin{aligned} \Delta_X &= 4A, \\ \Delta_M &= 8A + 16B + 8C_{\mathrm{shp}}, \\ \Delta_P &= 6A + 8B + \tfrac{16}{3}C_{\mathrm{shp}}, \\ \Delta_R &= 12A + 48B + 16C_{\mathrm{shp}} + \tfrac{16}{3}D, \end{aligned} \tag{29}$$

and they invert exactly, so the four shape coefficients are recoverable from four checkpoints. Δ_X contains A alone, which is why the axial cuts agree whatever the dispute does.

The dispute is two numbers and their difference, and the paper should print all three rather than describe them:

$$\begin{aligned} C_{\mathrm{old}} &= -\frac{211835444920651}{4405310420659200} = -0.0480863\ldots, \\ C_{\mathrm{new}} &= -0.020213328886166577, \\ \Delta_C &= C_{\mathrm{new}} - C_{\mathrm{old}} = 0.0278730\ldots \end{aligned} \tag{30}$$

The first is recorded as an exact rational and the second only as a float — that asymmetry is a fact about the two records, not an argument for either. The gap is 58% of $|C_{\mathrm{old}}|$, so this is not a precision disagreement that a tighter run would close.

Four further measurements locate the coefficient and bound where a disagreement could live. The distinction matters and is easy to lose: showing where a coefficient's *value* is concentrated is not by itself showing where a *discrepancy* can sit. Only the third item below does the second thing.

1. **The shape fit's C row sums to zero**, so no error in the Γ anchoring can move C_{shp} at all; the recorded run's own re-anchoring moved it by 4.6×10^{-15}, as the zero row sum requires. A scalar shift is translation-local and changes nothing observable, which is why the two anchored scalars $q^{(4)}_{\mathrm{band}}$ and $m^{(4)}_{\Gamma}$ are differently anchored coordinates and not rival estimates.

2. **The coefficient's value is concentrated, not spread.** Six of the 189 kernel records — the normal $(0, 0, 1)$ shell — carry 81% of the total weight, while the 120 rotation records carry 6% and are 255 times less sensitive per record. This says where the coefficient lives, not yet where the two records can differ.

3. **The agreed axial coefficient pins the normal sector.** All six normal records share one amplitude h, and it sets both shapes rigidly: $A_{\mathrm{normal}} = -h$ while the raw normal amplitude is $2h$, which in the shape normalisation of (25) is $C_{\mathrm{normal}} = h/2$, so

$$C_{\mathrm{normal}} = -\tfrac{1}{2} A_{\mathrm{normal}} \quad \text{exactly.}$$

   (The factor of four between $2h$ and $h/2$ is the raw-to-shape conversion, not a discrepancy.) Granting $A = 5/48$ — which both records share — together with the block decomposition, which puts $5/48 + 4x$ on the normal block and $-4x$ on the mixed one, therefore fixes h and with it 81% of the coefficient; and everything free to move without disturbing A totals less than half the disputed gap. A rival kernel must differ *structurally* in the A-carrying sector; no redistribution reaches the gap.

4. **The two kernels agree almost everywhere.** Their supports are identical and 144 of 189 records ride one common scale; the divergent cross sector is a single uniform ratio. The dispute is three amplitudes.

Read together these four sharpen the target and prefer no side of it. They say that a rival kernel must differ structurally in the A-carrying sector rather than by redistribution. That is a constraint on candidates built on the recorded 189-record support — which is where the measurement was made; a kernel on a different support is not constrained by it at all. Neither record is promoted here and neither is withdrawn: the interval of *values* between them is not narrowed. What is narrowed is the class of *kernels* that can reach either end, and that constraint bears on both records alike.

9.5 What cannot decide it

A large sealed sweep over 609 marked clusters has been the standing candidate for settling this. It cannot: its engine assembles coefficients through a Γ-point scalar and emits no kernel block at all, and by Theorem 35 Γ-point data are structurally incapable of constraining C_{shp}. This is a statement about the instrument, made by reading it, not a report of a failed run.

The instrument class that could eventually decide it now demonstrably exists: the sealed radius-two delivery of Section 5.4 emits a seed-space *matrix* at fifth order, not a scalar. What it lacks is off-axis momentum — a two-cube cluster has none — so it too cannot adjudicate. The requirement is now sharp on both counts: a decisive instrument must emit more than a Γ scalar *and* resolve an off-axis point such as P; the marked-cluster engine fails the first condition, the radius-two delivery the second, and a target-blind calculation satisfying both is exactly the next step named in the conclusion.

9.6 The tier collapse, as two integers

The two shape coefficients B and D vanish in the recorded kernel even though the two-hop enumeration generates both, and the degree count does *not* forbid them. The carrier numerator reaches a-degree three, because the projection contributes one further power of a: for any vector whose components have squared modulus a_i — the carrier is one, in whichever of the two equivalent labellings — the quadratic form $\mathrm{diag}(a_i^2)$ evaluates to $\sum_i a_i^3 = q^3 - 3qe_2 + 3e_3$, an explicit witness carrying a nonzero e_3. An earlier degree-bound mechanism proposed for the collapse was therefore retracted. What is checked is finer: every degree-three block amplitude is an integer multiple of one raw record weight x, and

$$B = 0 : \ (+1, +1, -2) \cdot x, \qquad D = 0 : \ (-3, +6, -3) \cdot x$$

over the displacement shells. The kernel does populate the forbidden tier; the dynamics cancels it with small integers. Any mechanism must reproduce those two vectors.

9.7 A near-Γ statement that does not adjudicate

One quantitative fourth-order statement can be made without adjudicating anything, because it depends on the kernel only through $\sqrt{W_4}$. Here

$$W_4 = \sup_k \bigl|\varepsilon_4(k) - \varepsilon_4(0)\bigr|$$

is the zone spread of the carrier-projected fourth-order correction about Γ, and $\theta \in (0, 1)$ is the fraction of the second-order isolation gap the fourth-order spread is allowed to occupy; we take $\theta = 1/2$ throughout. What the criterion needs is a lower bound on q over the region $|k| \geq r$, and there it is sharp: writing $q = \sum_m h(k_m^2)$ with $h(t) = 2(1 - \cos\sqrt{t})$, the identity $(s\cos s - \sin s)' = -s\sin s \leq 0$ makes h concave and increasing on $[0, \pi^2]$, so the minimum over the simplex sits at a vertex and

$$\min_{|k|\geq r} q(k) = 4\sin^2(r/2), \tag{31}$$

attained on a coordinate *axis*. With $\tau(u) \geq t_3 u^2$ for $u > 0$, the fourth-order shape terms are therefore dominated by the second-order dispersion outside a ball of radius $2\arcsin\bigl(\frac{u}{2}\sqrt{W_4/(\theta t_3)}\bigr)$, which is Ku to leading order with $K = \sqrt{W_4/(\theta t_3)}$: the two records give $K = 10.85$ and $K = 15.06$ — floating point, since one of the two bandwidths exists only as a float — and the larger bounds both. Jordan's inequality $q(k) \geq (4/\pi^2)|k|^2$ also holds on the whole zone and is tight at $k = 0$ and at the corner, but the corner is not where this bound is used; on the axis it is loose by $\pi^2/4$, and the constants it gives, 17.04 and 23.66, are exactly $\pi/2$ times the sharp ones. So the exclusion radius can be stated for either record without choosing between them. It is not a bound on the open coefficient: a value of C_{shp} outside the two records carries its own W_4, and nothing here confines the answer to the interval the records span. The statement is non-vacuous only for $u < 0.133$ at $\theta = 1/2$, and that threshold is the same for both constants — the sharp radius reaches π when $\frac{u}{2}\sqrt{W_4/(\theta t_3)} = 1$, which is exactly where $Ku = \pi$ for the Jordan constant, because the two bounds agree at the zone corner — and the excluded zone fraction grows as u^3.

10 External comparisons

External results are useful here when their provenance is independent of the present calculation. We use them only within the scope stated below; all normalisation conversions and coefficient matches are calculations of this paper.

Hamer 1989, both sectors. Let a_n denote the coefficient of x^n in Hamer's tabulated dimensionless mass series. The 1^{+-} rest-frame series agrees with E_{flat} at every shared order: $16/3 \div 2 = 8/3$ exactly, $a_1 = +1$, and $2a_2$, $4a_3$ matching $11/306$ and d_3 to 9.9×10^{-14} and 1.1×10^{-12}. Conditional on the unsigned-symbol identification, the 0^{++} series also matches the charge-even A_1^{++} Γ-point coefficients at x^2 and x^3, with the sign structure ($a_1 = -1$ in the scalar channel where the carrier's is $+1$) reproduced; the E^{++} doublet is not constrained by it, which is why Conditional result 29 uses the splitting rather than the level. The conversion $m_n = 2^{n-1}a_n$ is $x = 2u$ together with Hamer's normalisation $ma = (g^2/2)M$ — the 2^n from the substitution, the half from the normalisation — exactly, with no 4^r ambiguity anywhere. Hamer states in print that his x^3 and x^4 terms disagree with the earlier series of [3]. We report Hamer's comparison and do not independently adjudicate the earlier coefficient table.

The string tension. Let t_n denote the coefficient sequence in the transcribed Kogut–Pearson–Shigemitsu $SU(3)$ table [6], and let σ_n denote the corresponding ledger coefficient after the displayed convention conversion. The transcription satisfies exactly $2^{n-1}t_n = \sigma_n$ for $n = 2, \ldots, 5$, in the physical sign convention $\sigma_n^{\text{phys}} = (-1)^n \sigma_n^{\text{raw}}$. The same series reproduces E_{flat} term by term through u^3, including the discriminating u^3 coefficient.

Euclidean series. Historical Euclidean strong-coupling series by Münster, Seo and Smit [5, 7, 8, 9, 10] provide context only. The accompanying corpus contains a transcription comparison and an apparent eighth-order discrepancy between the 1982 erratum and the 1985 table. Because every original table page has not been independently re-transcribed for this paper, neither the apparent equality nor the discrepancy is used as evidence. No Hamiltonian coefficient is inferred from these Euclidean records.

The overlap obstruction. Schierholz [12] and Brandstaeter, Kronfeld and Schierholz [14] document the degradation of local-operator projection and its critical signal-to-noise scaling as the lattice spacing is reduced. Nothing quantitative is transferred to the present strong-coupling finite-order setting. Read only as a warning, these results motivate posing `G18` in a smeared basis; they supply no bound and do not identify the carrier with a physical glueball.

Cross-regime discipline. Two indexed papers declare an explicit scope firewall and neither supplies a value here: Davoudi–Raychowdhury–Shaw [22], whose Hilbert-space cost comparison is locked to 1+1 dimensions with matter, and Kadam–Raychowdhury–Stryker [23]. The rule is enforced by the literature validator rather than recorded in prose.

11 Scope

This work does not establish a four-dimensional infinite-volume Yang–Mills mass gap, a continuum glueball mass, convergence of the strong-coupling series to the continuum, regulator independence of the first dispersive order, the full $SU(3)$ fourth-order rest coefficient or off-axis shape, or an identification of any level here with a physical glueball. At Γ the three face orbitals transform as an axial vector of the cubic group, which with parity even and charge conjugation odd gives the retained-space label T_1^{+-}; this is an operator-space assignment, not a spectral one.

The carrier is macroscopically degenerate at finite L. Under the third-order factorisation input, its nearest incidence separation is $\Delta_L(u) = 4\tau(u)\sin^2(\pi/L) + O(u^4)$, which falls as L^{-2} and so provides no volume-uniform spectral isolation.

Three crossings are named explicitly, and none of them is made here.

- **Finite lattice to infinite volume** requires an inhomogeneous Wilson free-energy bound with a useful $\log K_\alpha$ and a source-radius reduction. Both are open, and are carried in the accompanying gap register as `G17`.
- **Protected level to particle** requires a volume-uniform overlap of a smeared, dressed T_1^{+-} operator built on the carrier with the transfer-matrix spectrum, and multi-plaquette survival of the protected level (`G18`). The published overlap measurement of Section 10 is why this theorem must live in the smeared basis.
- **Lattice to continuum** (`G19`) is untouched. Small u is strong coupling; the continuum is the opposite end of the axis.

All three crossings are required, none is established here, and nothing in this edition shortens that route.

Earlier editions carried three load-bearing prose premises here. Two of them are now proved — the retained shell (Lemma 15) and second-order process exhaustion (Lemma 16) — with their finite combinatorics machine-enumerated and their two classical representation-theoretic inputs named in the proofs. Those proofs are new in this edition and have not yet faced external referees; a reader auditing this paper should start there. One premise remains.

That the complete linked charge-even Bloch symbol is the unsigned incidence $\mathcal{N}(k)$ — the signed incidence with the boundary orientations dropped — is not derived. The cubic and range in Section 6 are exact algebraic properties of the defined comparison operator; only their physical charge-even interpretation, bandwidth and Γ-splitting rest on this additional hypothesis.

12 Evidence, authority, and reproducibility

The accompanying dependency-free program `verify_publication_core.py` uses exact integers and `Fraction` arithmetic. It checks the all-rank coefficient ledger and Weingarten sums, signed incidence and torus homology, closed-cube shell, two-cube connected geometry and reported channel arithmetic, the unsigned cubic, the saved historical Hodge pencil, the fourth-order obstruction, the coupling convention, the planar closed form (18), and the exhaustive shared-link geometry, cycle censuses and torus ranks through $L = 5$ — thirty-eight checks, every value a `Fraction` and no float constructed anywhere in the file. A second program, `verify_radius2_report.py`, holds everything the sealed radius-two artifact of Section 5.4 can support: it is also standard-library only (the payload is read with `zipfile` and `struct`, and the 11×11 block is diagonalised by cyclic Jacobi), but it works in floats at stated tolerances, and the two files are kept separate precisely so that the exact one stays float-free. Neither reconstructs the finite-precision Clebsch–Gordan inputs, replays the third-order microscopic contractions, or chooses C_{shp}.

Every sentence of this paper that a machine check certifies carries an inline `\chk` marker in the source; the markers are extracted verbatim into Appendix G, so the coverage claim is itself auditable rather than asserted.

12.1 Claim-level evidence map

12.2 Upstream corpus and downstream verifier

The organised `ALL THEORY` archive was consulted as the upstream scientific corpus. Its four-document v4.3 authority stack is frozen by byte count and SHA-256 in `export/MAN_GOV_export_manife`

Claim	Status	Evidence in this edition
Signed incidence spectrum and L^3+2 carrier	Proven	Analytic integer topology; exact replay
Adjacent all-rank coefficient t_N	Proven locally	Weingarten derivation; exact replay
Global order-u^2 torus assembly	Proven (new this edition)	Lemmas 15, 16; machine-enumerated combinatorial cores; not yet externally refereed
Two-cube six-channel census	Reported numerical	Rational reconstructions; shared amplitude lineage; assembly test
Unsigned-incidence cubic and range	Proven algebraically	Exact characteristic polynomial and finite-grid replay
Physical charge-even identification	Conditional	Complete linked symbol, including vacuum route, not derived
$SU(3)$ factorisation through u^3	Upstream cold-exact	251/251 exact gates reported by the frozen authority; raw replay not bundled here
Historical fourth-order pencil	Exact for saved kernel	Hash-pinned payload and cold fixed-kernel checks
Complete physical fourth-order kernel	Disputed/open	Historical and August candidates differ off axis
Radius-two rational sub-block and C_4/C_5 increments	Reported numerical	Sealed hash-pinned artifact; stdlib float verifier at stated tolerances
Infinite-volume particle or continuum gap	Open	No uniform overlap, full-sector, or continuum bridge

Table 1: Truth status and evidence level are separate. "Exact for a saved kernel" does not establish that kernel's physical identification.

`st.csv`; all four entries were rehashed for this edition and matched. The downstream verifier is WORKHOUSE. Document flow is one-way—frozen corpus to hash manifest to verifier—while verifier findings return to review and status records. The later two-cube results are not in the August 20 corpus and are therefore kept as later WORKHOUSE-local evidence.

The current hostile-scope audit branch is pinned at `ff9a5976a5b03cd763e3e50daf43261387 70460d`. On Windows, its registry was rerun locally: 229/230 registered checks passed. The sole failure was the known path-separator comparison for a restored payload; the payload hash and all paper-relevant arithmetic checks passed. This is a platform-specific local audit result, not a claim that the complete test suite or Lean tree was rerun. The source-level `\chk` labels remain in this TeX file but are suppressed typographically; Appendix G prints all of them, so the coverage claim is auditable rather than asserted. The label guard now reads every edition in `paper/`, this one included, and it resolves all but twelve of the labels below against the live registry. Those twelve are recorded as a standing finding rather than quietly dropped: each sits under a displayed result and promises a command that currently returns nothing, and the fix is to register the twelve checks, not to stop printing the twelve labels.

The authority files are identified by the following SHA-256 values:

- master v4.3: `1f8295c3a797aeb854b4b0ca82485bd7f4a149617777d2a43fc147a4a53c26dd`;
- detailed formulas v3.1: `924d5b26f865e33f530d166735e847b89f60df7fe363995e2788ab24 e1ff4254`;
- consolidation guide v4.3: `60b8a93c8149049f50ed3ce8e0cfd1b379723b6765f201395a2fda 6c4a7e7aa7`;
- canonical source manifest v4.3: `730bc99ed8d749de0dffbabfda85cd4512d8f7748782620b69`

`4759a03bacde05`.

12.3 Data and code availability

This release directory contains the TeX source, generated vector figures, figure generator, exact verifier, build log, rendered-page audit, and a SHA-256 manifest. The historical kernel payload used in Reported result 34 has SHA-256 `635d40fa8a5d7da841fd30f36185eb96f14ec4c040678ddd8fb010379afb2900`. The pinned `ymcirc` $d = 3$, $\mathcal{B} = 4$ data blob is identified in Ref. [28] and has SHA-256 `4cad9538f2c052488556362e0cf6e4e63363e7f5801597aeab9013ef945c59dc`. The exact stranded-flux audit source and result have SHA-256 `a42800a91809a77bbd1d339ac604ea80a234eed305377bbabbf890fee98fde55` and `d215fa1e9cd0bde0410eec16c687182c30bc62fa3f28c1445fd81e93888e1d60`. The sealed cutoff-free radius-two spectrum of Section 5.4 and its certificate are bundled, with SHA-256 `52a17f87d3990b234635828a0c92f73f781e9a3abec8bb3c127e990b7a717e6c` and `b9b14d28438585fc4f7d88f96497e9c6e01d0532a8402ffd6db17258a1638c0e`; `verify_radius2_report.py` refuses to run against any other bytes. The two-cube build and its finite-precision Clebsch–Gordan payloads are not reproduced by the compact verifier; the exact statements that depend on them are labelled reported. A durable tag or archival DOI is still needed before journal submission.

Machine checks certify algebra relative to their definitions and inputs. They do not replace derivations, promote numerical contractions to symbolic ones, or certify an uncompleted off-axis contraction.

13 Conclusion

The strongest result is no longer only local. For every $N \geq 3$, the four shared-link fusion channels and their resolvent gaps give the closed coefficient t_N, the oriented cellular boundary fixes its sign and geometry, and — with the two completeness premises now proved (Lemmas 15 and 16) — the assembly into a homological flat carrier of dimension $L^3 + 2$ is a theorem: the flat sector is not a fine-tuned interference band but the cycle space of the lattice two-complex, degenerate because $\partial_2 \partial_3 = 0$. Those two proofs are the newest and least-reviewed material here, and they deserve the reader's hardest look. The $SU(3)$ two-cube calculation supplies a nontrivial operator-level assembly check, including channel-by-channel geometry and orbit localisation, but shares its amplitude lineage and remains numerical at the contraction layer.

The unsigned-incidence cubic is exact as an algebraic comparison; its physical charge-even identification is not assumed silently. At fourth order the saved historical kernel has an exact positive generalized Hodge pencil and stencil, yet the later candidate differs in an off-axis coefficient that no scalar reanchoring, Γ datum, or axial datum can determine. The decisive next object is therefore one target-blind, hash-sealed calculation that returns the rest scalar and off-axis shape from the same kernel. Until then the finite-order operator theorem is substantial, but the physical fourth-order kernel, infinite-volume particle interpretation, and continuum limit remain open.

A Channel weights in closed form

Substituting (10) into (14),

$$w_{\mathbf{1}} = -\frac{2}{N(N^2-1)}, \qquad w_{\Lambda^2 F} = -\frac{N-1}{(N+1)(2N-3)},$$
$$w_{\mathrm{Adj}} = -\frac{2(N^2-1)}{N(2N^2-1)}, \qquad w_{\mathrm{Sym}^2 F} = -\frac{N+1}{(N-1)(2N+3)}.$$

Their family sums are (15) and (16), and the difference has numerator $2N(N^2-4)$, which is the $SU(2)$ zero and the positivity for $N \geq 3$ at once. At $N = 3$: $w_{\mathbf{1}} = -1/12$, $w_{\mathrm{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$, $w_{\mathrm{Sym}^2 F} = -2/9$, which are the four values Reported result 18 matches against (22).

B The Weingarten index sums

With $\mathrm{Wg}(e) = 1/(N^2-1)$ and $\mathrm{Wg}((12)) = -1/(N(N^2-1))$, and σ, τ running over S_2,

$$\begin{aligned}&\langle U_{i_1 j_1} U_{i_2 j_2} \bar{U}_{i'_1 j'_1} \bar{U}_{i'_2 j'_2} \rangle \\ &\quad = \sum_{\sigma,\tau} \mathrm{Wg}(\sigma\tau^{-1}) \prod_a \delta_{i_a i'_{\sigma(a)}} \delta_{j_a j'_{\tau(a)}}.\end{aligned}$$

Summing the delta products over all N^4 index quadruples,

$$M_{\mathrm{direct}} = \mathrm{Wg}(e)(N^4 + N^2) + 2\,\mathrm{Wg}((12))N^3 = N^2,$$

$$M_{\mathrm{cross}} = 2\,\mathrm{Wg}(e)N^3 + \mathrm{Wg}((12))(N^4 + N^2) = N.$$

Both are confirmed by explicit summation at $N = 3, 4, 5$. The related fourth moment $\int |U_{11}|^4 \, dU = 2/[N(N+1)]$ is $1/6$ at $N = 3$ and in particular nonzero, which is what refuted a claimed Haar vanishing in the balanced $(2, 2)$ sector.

C The $SU(3)$ coefficient ledger

Order	Quantity	Value
u^0	electric plaquette	$8/3$
u^1	scalar	1
u^2	$BB^\dagger$ coefficient	$5/612$
u^2	flat scalar	$11/306$
u^3	$BB^\dagger$ coefficient	$1975/124848$
u^3	per-neighbour leakage	$-12331/249696$
u^3	flat scalar	$-109151/249696$

The u^1 row is $SU(3)$-only rather than an $SU(3)$ specialisation: it is $-\langle p, -|V|p, -\rangle = 1$, which needs the ϵ channel and vanishes for $N \geq 4$. Of the rest, the u^0 row and the u^2 $BB^\dagger$ coefficient specialise all-rank statements ($2C_F$ and t_N); the u^2 flat scalar inherits the $SU(3)$ tower term through (23); and the three u^3 rows are $SU(3)$-only and conditional on the supplied domino ledger, exactly as Section 7 says.

D Unsigned comparison and conditional charge-even ledger

Conditional on identifying the unsigned-incidence comparison with the complete linked charge-even symbol, the same convention, with $\lambda \in \{12, -4\}$ and (23), gives:

Order	Quantity	Value
u^2	hopping $t_{2,+} = \ell_3$	$-11/306$
u^2	on-site diagonal	$223/1020$
u^2	band bottom ($\lambda = 12$)	$-217/1020$
u^2	band top ($\lambda = -4$)	$1109/3060$
u^2	bandwidth $16\lvert t_+\rvert$	$88/153$
u^3	hopping $t_{3,+}$	$-6335/249696$
u^3	on-site diagonal ($\lambda = 0$)	$52163/260100$
u^3	band bottom ($\lambda = 12$)	$-54049/520200$
u^3	band top ($\lambda = -4$)	$471353/1560600$

Because $t_{r,+} < 0$ at both orders, the $\lambda = -4$ endpoint is the band *maximum* in this sector. A certificate key naming it `bandmin` at both orders is a naming error against its own arithmetic, recorded rather than silently corrected.

E Finite-volume chain calculation

For $i < j$,

$$\partial_2[x; i, j] = [x + e_i; j] - [x; j] - [x + e_j; i] + [x; i],$$

$$\begin{aligned} \partial_3[x; 1, 2, 3] = [x + e_1; 2, 3] &- [x; 2, 3] \\ &- [x + e_2; 1, 3] + [x; 1, 3] \\ &+ [x + e_3; 1, 2] - [x; 1, 2]. \end{aligned}$$

Substituting the second into the first cancels every edge pairwise, so $\partial_2\partial_3 = 0$ over $\mathbb{Z}$. On the torus $\ker \partial_3$ is generated by the sum of all oriented cubes, so $\operatorname{rank} \partial_3 = L^3 - 1$.

F The two-cube geometry

The eleven oriented plaquettes of the $(3, 2, 2)$ prism, by base vertex and ordered plane axes, are

$$\begin{aligned} &(000; 12),\ (000; 13),\ (000; 23),\ (001; 12), \\ &(010; 13),\ (100; 12),\ (100; 13),\ (100; 23), \\ &(101; 12),\ (110; 13),\ (200; 23), \end{aligned}$$

with left, right and shared-face index sets $I_L = (0, 1, 2, 3, 4, 7)$, $I_R = (5, 6, 7, 8, 9, 10)$, $I_F = (7)$. Building $G = BB^\mathsf{T}$ from the oriented boundaries and applying the geometric Möbius fold (20) leaves exactly four nonzero off-diagonal pairs, $(0, 5), (1, 6), (3, 8), (4, 9)$, each equal to -1: four disjoint 2×2 blocks and three isolated directions, which is why the spectra in Section 5 are closed form once the diagonals below are supplied. The two connected diagonals are

$$D_{\mathcal{B}=6} = \tfrac{1}{612} \operatorname{diag}(-2317, -2317, -2295, -2317, -2317,$$

$$-2317, -2317, 0, -2317, -2317, -2295),$$

$$D_{\mathcal{B}=4} = \mathrm{diag}(-\tfrac{7}{4}, -\tfrac{7}{4}, -\tfrac{15}{4}, -\tfrac{7}{4}, -\tfrac{7}{4},$$

$$-\tfrac{7}{4}, -\tfrac{7}{4}, 0, -\tfrac{7}{4}, -\tfrac{7}{4}, -\tfrac{15}{4}),$$

so both spectra are one line of arithmetic: a pair block contributes $D_{ii} \mp c$ twice and each isolated direction its own D_{ii}. The $\mathcal{B} = 4$ comparator on the same geometry gives 0 once, $-\frac{15}{4}$ twice, $-\frac{11}{6}$ four times, and $-\frac{5}{3}$ four times: the paired levels sit *above* the isolated $-15/4$ pair, where at $\mathcal{B} = 6$ they sit below it. The three isolated levels $\{-\frac{15}{4}, -\frac{15}{4}, 0\}$ are identical in the two truncations, so the entire effect of the cutoff lives in the four cross-cell blocks. That reversal is not the coefficient's sign by itself — within a 2×2 block the pair splits as $D \mp c$, which is invariant under $c \to -c$ — but the truncation's effect on the connected diagonal as well. What the sign of c alone does reverse is the ordering on a *single* cube, where the kernel is $\alpha I + tG$ with one diagonal and the spectrum of G fixed; that is the shell reversal the one-cube reconstructions report.

G Machine-check coverage

The paper's discipline is that a machine check, not a sentence, is the unit of evidence. This appendix makes that auditable: every inline `\chk` marker in the source — each one naming the check that certifies the sentence it sits beside — is extracted verbatim by `make_coverage.py` and printed here. A reader can therefore see exactly which statements are covered, and, by their absence from this list, which are not: the remaining load-bearing hypothesis of Section 11 — the charge-even identification — appears nowhere below, and that is the point.

The source carries 127 inline markers. Each line below is one marker, verbatim, under the section that carries it.

Introduction

- the discrete index theorem gives 0 on the signed face-edge operator, bounding nothing

Hamiltonian and projected sector

- the printed towers are canonical-u: 4*Delta(3u/2) reproduces them verbatim
- the 4**r rescaling breaks the bridge: order 2 off by 16, order 3 by 64
- at N = 2 the C-odd hopping vanishes while the unsigned-channel continuation does not

Oriented incidence and the exact carrier

- B B^dagger = q I - d conj(d)^T for the curl incidence
- dim Z_2 = L^3 + 2 by rank, not by re-arranging the formula
- cube boundaries and three wrapping sheets SPAN Z_2
- the L^3+2 count is chain-level, not the Bloch convention
- the Bloch and chain routes to the carrier agree
- the sheets average to harmonic representatives, and the Gamma block IS b_2

The wrapping sheets are cycles, not harmonic — and their harmonic representatives

- FINDING: the wrapping sheets are cycles but NOT harmonic
- the sheets average to harmonic representatives, and the Gamma block IS b_2

Exact adjacent shared-link dynamics and conditional assembly

- each channel gap is C_F + C_R/2, and the weights sum to one
- each channel gap is C_F + C_R/2, and the weights sum to one

The channel weights are a theorem

- the shared-link weights are Weingarten, not an isotropy assumption
- the Weingarten route is independent of the corpus
- the published dimension-ratio matrix element is this registry's weight formula
- the four channel weights follow from dimension and Casimir
- A_N and B_N are the channel sums, not transcriptions

The shared-link law

- t_N = B_N - A_N
- t_N > 0 for N >= 3
- two distinct torus faces share at most one link, exhaustively at L = 3 and 4
- t_2 = 0 and t_3 = 5/612
- large-N expansion of t_N through 1/N^9
- 1 - 4N^3 t_N has the closed form of eq. (planar), positive for all N >= 2
- 4N^3 t_N increases monotonically to 1: shifted derivative coefficients are nonnegative

The vacuum-mediated route

- ell_N = A_N + B_N + 1/C_F, the vacuum-mediated route at every rank

Closing the two premises

- the torus graph is simple, and the only short cycles are the L = 3 winding triangles
- every 4-cycle is an elementary face, except exactly the 3L^2 straight winding loops at L = 4
- two distinct torus faces share at most one link, exhaustively at L = 3 and 4

Finite-volume width

- the zone maximum of q is 12 only at even L
- q_min on the L-torus grid is 4 sin^2(pi/L)

The orientation signs are essential

- the two band spans ARE the two incidence spectra

The second-order coefficient on a two-cube space

- connected first order vanishes by inclusion-exclusion, not by cancellation of numbers
- the connected two-cube geometry has exactly four cross-cell pairs, each -1
- the B=6 connected kernel is (5/612) G_conn + diag, with the certified spectrum
- the certificate's own gates pass, target-blind, with the wrong-sign control rejected
- the B=6 six-channel census sums to the registry's own t_3 = 5/612
- the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel
- every shared-link channel is separately proportional to the geometry, on all 56 pairs

One truncation, two constructions

- B=6 retains every adjacent shared-link channel; B=4 provably cannot
- the retention rule is C2(rho) + 2 C2(3) <= B, and both retentions it decides are equalities
- the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel
- FINDING: the T1 link cutoff reverses the sign of t_3, and 14/153 is what it omits
- FINDING: a full T1 = B = 4 cube Hamiltonian reproduces -1/12 and the reversed shell
- FINDING: the B = 6 cube flips the sign back to +5/612, closing the decisive test
- the B = 6 scalar misses the bridge's by exactly the same-face sextet route

What the cutoff moves, and where

- the connected diagonal is orbit-constant, and only the transporting orbit moves

The one-cube shell, and why its flat level cannot move

- the one-cube shell is A1 + T1 + E, and its flat level is the S^2 fundamental class
- the certificate's finite-volume fingerprints, and 29 = L^3 + 2 is the Lean cycle count

A cutoff-free radius-two witness at finite coupling

- the sealed rational sub-block straddle is exactly 2 t_3 = 5/306
- six sealed eigenvalues match the three rationals to 1e-12, by a stdlib Jacobi diagonalisation
- direct C4/C5 matched-gap coefficients agree with their weak-grid intercepts to 5e-6
- the fifth-order seed shape is traceless with nonzero trace-square: the instrument emits more than a scalar

The unsigned-incidence comparison operator

- the unsigned-incidence characteristic polynomial is mu(mu - p)^2 = 4 a_1 a_2 a_3
- p + q = 12: one zone function runs both sectors
- the Bloch cubic IS the finite L = 3 and L = 4 plaquette spectrum, exactly
- the C-even range [-4, 12] is exact; the top only at Gamma, the floor on three planes
- the C-even band touches its floor exactly on the three planes k_j = pi
- the C-even band touches its floor exactly on the three planes k_j = pi
- the C-even spectra at the four high-symmetry momenta

Conditional charge-even assembly

- one assembly formula gives every registered band value
- one assembly formula gives every C-even value at both orders
- the plaquette graph is 12-regular and two faces share at most one link
- both declared coincidences, checked: one is ell_N at all ranks, the other is bare

Where the rest frame is not blind

- the C-even Gamma point pins t_+, exactly where no C-odd Gamma datum can
- Schur alone fixes both Gamma blocks: the C-odd triple is irreducible, the C-even one is not
- the C-even curvature is (4/3)|t_+| at both orders, and isotropic
- the C-even bandwidth is 16|t_+| at every order; the C-odd manifold width is 12|t_-|
- at N = 2 the C-odd hopping vanishes and the C-even one does not

$SU(3)$ through third order

- d_3 = 7/32 + 12*leak_3 - 4*b_3
- leak_3 is assembled from the domino diagonal and the vacuum piece
- E_flat and t(u) carry the ledger coefficients
- d_- = 1/2 + 12*leak_2, and leak_2 = -11/306
- FINDING: no Gamma-point datum can constrain the hopping

What the homology protects

- a boundary-factorised correction shifts the carrier without dispersing it, for generic M
- the carrier projection is where the 1/q comes from
- the shape family is not symmetry-complete: cubic symmetry alone permits q^2 as well
- clearing the denominator reproduces the five-element numerator basis

An exact theorem for the saved historical kernel

- v10a.26 A, B, D match the sealed rationals within 2.3e-13
- FINDING: alpha_new falls outside the corpus's own 2.3e-13 bound

The obstruction is polynomial

- FINDING: the retained Gamma/axis data cannot identify C_shp
- Phi_C vanishes at Gamma along every direction
- the crosswalk is exactly scalar on the momentum axes
- on an axial cut the mixed invariants vanish and the norm divides

A second witness

- FINDING: an explicit second witness C_alt exhibits the C2 non-identifiability
- the C_alt witness IS the balanced eps-free continuation, and Delta_C/(A/2) = 15/32

Where the coefficient lives

- checkpoint values at X, M, P, R
- the four extraction formulas invert the ansatz
- X is blind to B, C, D — it fixes A alone
- the shape fit's C row sums to zero, so no Gamma-anchor error can move C_shp at all
- a translation-local scalar shift changes nothing observable
- FINDING: C_shp is carried by 6 of 189 records, not spread across the kernel
- FINDING: A = 5/48 pins the normal sector's whole C4 contribution
- C_normal = -A_normal/2: the agreed axial coefficient pins the normal channel
- the two 189-record kernels agree everywhere except three amplitudes, and the on-site anchor swap moves C by exactly zero

What cannot decide it

- FINDING: the marked-cluster engine emits the Gamma scalar only — a completed 609-sweep cannot decide C_shp

The tier collapse, as two integers

- RETRACTED: the vertex count does NOT forbid B_shp and D_shp
- FINDING: the tier collapse is two integer cancellations at record level
- the vanishing coefficients are exactly the degree-3 ones

A near-Γ statement that does not adjudicate

- Jordan bound q(k) >= (4/pi^2)|k|^2 holds on the whole zone
- the exact zone minimum of q on |k| >= r is 4 sin^2(r/2), and Jordan overstates it by pi/2
- the criterion survives C2: K depends on the kernel only through sqrt(W_4)
- the statement is non-vacuous only below an explicit coupling

External comparisons

- Hamer's 1+- series matches the C-odd Gamma-point coefficients through x^3
- Hamer's 0++ series matches the C-even Gamma-point coefficients through x^3
- the m_n = 2^(n-1) a_n bridge is the x = 2u conversion
- the Hamer table metadata is pinned; no fourth-order agreement is used
- the KPS 1980 string-tension table equals the certified sigma series EXACTLY
- sigma_n^phys = (-1)^n sigma_n^raw, and C5 is the n = 3 case
- the ratio and sigma series reproduce E_flat exactly
- the errata-resolved Euclidean series is doubly sourced, transcription for transcription
- FINDING: Munster's 1985 table shifts his 1982 erratum at eighth order
- the overlap obstruction was published in 1988, and it scales
- a cross-regime paper never supplies a value

Scope

- Delta_L = 4 tau(u) sin^2(pi/L) is positive and falls as L^-2

Upstream corpus and downstream verifier

- FINDING: the v2 publication edition prints twelve labels that name no check

Channel weights in closed form

- the four channel weights follow from dimension and Casimir

The Weingarten index sums

- the shared-link weights are Weingarten, not an isotropy assumption
- SU(3) Weingarten values follow from the general formula
- the fourth moment integral |U_11|^4 = 1/6 at N = 3

The $SU(3)$ coefficient ledger

- the manuscript's SU(3) ledger is this registry, value by value

Unsigned comparison and conditional charge-even ledger

- one assembly formula gives every C-even value at both orders
- FINDING: the certificate key 'bandmin' holds the band MAXIMUM, at both orders

Finite-volume chain calculation

- d_2 d_3 = 0 on the built complex
- rank d_3 = L^3 - 1 on the built complex

The two-cube geometry

- the B=4 comparator on the same geometry gives -1/12 with the reversed spectrum

References

[1] B. Eckmann, *Harmonische Funktionen und Randwertaufgaben in einem Komplex*, Comment. Math. Helv. **17**, 240–255 (1944), doi:10.1007/BF02566245.

[2] J. B. Kogut and L. Susskind, *Hamiltonian formulation of Wilson's lattice gauge theories*, Phys. Rev. D **11**, 395–408 (1975), doi:10.1103/PhysRevD.11.395.

[3] J. B. Kogut, D. K. Sinclair and L. Susskind, *A quantitative approach to low-energy quantum chromodynamics*, Nucl. Phys. B **114**, 199–236 (1976), doi:10.1016/0550-3213(76)90586-1.

[4] D. Weingarten, *Asymptotic behavior of group integrals in the limit of infinite rank*, J. Math. Phys. **19**, 999–1001 (1978), doi:10.1063/1.523807.

[5] G. Münster, *Strong coupling expansions for the mass gap in lattice gauge theories*, Nucl. Phys. B **190**, 439–453 (1981), with addendum **200**, 536–538 (1982) and erratum **205**, 648 (1982), doi:10.1016/0550-3213(81)90570-8.

[6] J. B. Kogut, R. B. Pearson and J. Shigemitsu, *The string tension, confinement and roughening in $SU(3)$ Hamiltonian lattice gauge theory*, Phys. Lett. B **98**, 63–68 (1981), doi:10.1016/0370-2693(81)90369-5.

[7] K. Seo, *Glueball mass estimate by strong coupling expansion in lattice gauge theories*, Nucl. Phys. B **209**, 200–216 (1982), doi:10.1016/0550-3213(82)90110-9.

[8] C. J. Smit, *Estimate of glueball masses from their strong coupling series in lattice QCD*, Nucl. Phys. B **206**, 309–320 (1982), doi:10.1016/0550-3213(82)90537-5.

[9] G. Münster, *Physical strong coupling expansion parameters and glueball mass ratios*, Phys. Lett. B **121**, 53–55 (1983), doi:10.1016/0370-2693(83)90201-0.

[10] G. Münster, *Effective transfer matrix for low-lying glueball states in lattice gauge theory*, Nucl. Phys. B **256**, 67–76 (1985), doi:10.1016/0550-3213(85)90385-2.

[11] B. Sutherland, *Localization of electronic wave functions due to local topology*, Phys. Rev. B **34**, 5208–5211 (1986), doi:10.1103/PhysRevB.34.5208.

[12] G. Schierholz, *Glueball masses in $SU(3)$ lattice gauge theory*, Nucl. Phys. B Proc. Suppl. **4**, 11–20 (1988), doi:10.1016/0920-5632(88)90075-8.

[13] C. J. Hamer, *Hamiltonian strong coupling expansions for glueball masses in $SU(3)$*, Phys. Lett. B **224**, 339–342 (1989), doi:10.1016/0370-2693(89)91242-2.

[14] F. Brandstaeter, A. S. Kronfeld and G. Schierholz, *Critical signal-to-noise ratio and glueball mass calculations*, Nucl. Phys. B **345**, 709–729 (1990), doi:10.1016/0550-3213(90)90406-4.

[15] A. Mielke, *Ferromagnetism in the Hubbard model on line graphs and further considerations*, J. Phys. A **24**, 3311–3321 (1991), doi:10.1088/0305-4470/24/14/018.

[16] J. Carlsson and B. H. J. McKellar, *$SU(N)$ lattice gauge theory on a single cube*, arXiv:hep-lat/0303022 (2003).

[17] B. Collins and P. Śniady, *Integration with respect to the Haar measure on unitary, orthogonal and symplectic group*, Commun. Math. Phys. **264**, 773–795 (2006), doi:10.1007/s00220-006-1554-3.

[18] D. L. Bergman, C. Wu and L. Balents, *Band touching from real-space topology in frustrated hopping models*, Phys. Rev. B **78**, 125104 (2008), doi:10.1103/PhysRevB.78.125104.

[19] T. Zaslavsky, *Matrices in the theory of signed simple graphs*, in *Advances in Discrete Mathematics and Applications: Mysore, 2008*, Ramanujan Mathematical Society Lecture Notes Series **13**, 207–229 (2010), arXiv:1303.3083.

[20] J.-W. Rhim and B.-J. Yang, *Classification of flat bands according to the band-crossing singularity of Bloch wave functions*, Phys. Rev. B **99**, 045107 (2019), doi:10.1103/PhysRevB.99.045107.

[21] A. J. Kollár, M. Fitzpatrick, P. Sarnak and A. A. Houck, *Line-graph lattices: Euclidean and non-Euclidean flat bands, and implementations in circuit quantum electrodynamics*, Commun. Math. Phys. **376**, 1909–1956 (2020), doi:10.1007/s00220-019-03645-8.

[22] Z. Davoudi, I. Raychowdhury and A. Shaw, *Search for efficient formulations for Hamiltonian simulation of non-Abelian lattice gauge theories*, Phys. Rev. D **104**, 074505 (2021), doi:10.1103/PhysRevD.104.074505.

[23] S. V. Kadam, I. Raychowdhury and J. R. Stryker, *Loop-string-hadron formulation of an $SU(3)$ gauge theory with dynamical quarks*, Phys. Rev. D **107**, 094513 (2023), doi:10.1103/PhysRevD.107.094513.

[24] K. Roychowdhury, J. Attig, S. Trebst and M. J. Lawler, *Supersymmetry on the lattice: Geometry, topology, and flat bands*, Phys. Rev. Research **6**, 043273 (2024), doi:10.1103/PhysRevResearch.6.043273.

[25] K. T. K. Chung and M. J. P. Gingras, *2-form $U(1)$ spin liquids: A classical model and quantum aspects*, Phys. Rev. B **111**, 064417 (2025), doi:10.1103/PhysRevB.111.064417.

[26] A. N. Ciavarella, I. M. Burbano and C. W. Bauer, *Efficient truncations of lattice gauge theory for quantum simulation*, Phys. Rev. D **112**, 054514 (2025), doi:10.1103/ylqb-phv5; arXiv:2503.11888v5.

[27] P. Balaji, C. Conefrey-Shinozaki, P. Draper, J. K. Elhaderi, D. Gupta, L. Hidalgo and A. Lytle, *Perturbation theory, irrep truncations, and state preparation methods for quantum simulations of $SU(3)$ lattice gauge theory*, Phys. Rev. D **113**, 094505 (2026), doi:10.1103/m719-7tdf; arXiv:2509.25865v3.

[28] HEPQIS–UIUC, *`ymcirc`: lattice-gauge-theory circuit and matrix-element tools*, GitHub repository, commit `e9e190bfda405608de9cab71c0df0161cfcb1a10`, data blob `1808849c74b36f8082e7030d074681be4bb8c0fd` (accessed 28 August 2026), repository; data introduced in PR 76.

[29] T. Hazra, *New class of exactly flat topological bands—compact localised states protected by local graph topology*, arXiv:2607.22831v1 (2026).